OrthoHMM: HMM-Centered Group Inference With Phylogenetic Refinement

Working manuscript with reporting-scope amendments approved on October 12, 2026 after the retained results were available. This changes interpretation and completion scope, not the method, scores, endpoints or frozen protocols. Four admitted native QfO cells are unchanged; later failures supply no score. Preserves the preceding manuscript and terminal-failure reporting. Publication completion remains subject to the remaining deliverable audit; this reporting revision alone is not proof of readiness or archival deposition.

Abstract

Orthology inference requires balancing homolog recovery against separation of paralogs. We evaluated an HMM-centered pipeline with candidate-family expansion and phylogenetic refinement against established orthology tools. QfO and OrthoBench were the primary development-exposed benchmarks; a frozen YGOB evaluation assessed transfer to additional taxa. Phylogenetic OrthoHMM improved on its high-sensitivity configuration but did not consistently outperform full OrthoFinder. On OrthoBench, their group-recovery F1 values were 74.11% and 72.74%, respectively. On corrected QfO, OrthoFinder had higher VGNC, SwissTrees and TreeFam-A point estimates, whereas OrthoHMM had higher GO/EC similarity and FAS. A matched-recall simulation control supported an initial HMM-search contribution in a fixed graph procedure, not full-pipeline superiority. Novel-taxon transfer does not establish family-disjoint generalization. VGNC and GO/EC use conventional native error bars; pooled-reference TreeFam is descriptive supporting evidence. FAS ranges address conditional sampling, and the project-defined six-metric mean remains descriptive. These limits do not establish statistically supported overall superiority. The completed shared-host matched-resource panel retains all 27 attempts, with 24 eligible measurements; it does not establish isolated comparative efficiency. The evidence supports a bounded contribution and an explicit precision-recall trade-off, not universal accuracy or efficiency claims. Fresh native ablations retain neutral downstream-profile effects and a SwissTrees precision-recall trade-off whose adjusted F1 interval includes zero. Higher native GO/EC means reflect scored-pair membership differences, not changed serialized similarities on shared pairs. Synthetic null scoring revealed composition-dependent behavior of the approximate significance filter. Heterogeneous-length simulations favored full OrthoFinder; supplied generating trees did not establish an accuracy gain, and stronger topology perturbations reduced F1 and recall in several conditions. A fixed-candidate generating gene-tree diagnostic localized most divergent-condition recall loss upstream of reconciliation; residual event-history traces identified satellite-constraint losses and duplication evidence absent from retained candidates.

Methods

The retained configurations are high-sensitivity OrthoHMM and the satellite_v2 phylogenetic pipeline. The original OrthoHMM preprint (Steenwyk et al. 2024) describes its lineage, not all subsequent implementation changes. The prospective method was frozen at `7f3a9e4`, with BLOSUM62, E-value threshold 1e-4, Leiden CPM resolution 0.1 and seed 4 (Traag, Waltman, and van Eck 2019; Traag, Van

Dooren, and Nesterov 2011). The phylogenetic configuration expands candidate families, infers gene and species trees, and applies positive-paralogy pair inference. Historical runs retain their actual revisions rather than inheriting this prospective pin. The method diagram distinguishes initial search, profile refinement, candidate expansion and phylogenetic inference.

Initial search uses banded local maximum-path scoring with match, insert and delete states, integer BLOSUM62 emissions and fixed transition costs. Raw scores receive an approximate significance filter before geometric-mean length normalization. MSA-derived profiles learn match emissions, while retaining uniform transition costs; strict expansion requires both a member-score threshold and an initial sequence-supported anchor. This is not demonstrated HMMER/phmmer equivalence or calibrated assignment confidence. The numerical specification records the recurrence, constants, profile construction and acceptance rules at the full frozen source revision. No scoring/default change accompanies it.

A prespecified null-score protocol examined that significance filter without fitting parameters. Three residue compositions and lengths 50/150/400 each used ten seeds and 1,000 independently drawn query-target pairs per seed. Each pair had a one-target database and was scored with full and width-64 bands: 90,000 independent pairs, not 180,000 independent observations. Five fixed cutoffs yielded 90 tail endpoints with Bonferroni-adjusted exact binomial intervals. These intervals describe the synthetic generator, not reference-family or orthology uncertainty.

Comparators were OrthoFinder 3.1.5 (David M. Emms et al. 2026a, 2026b), SonicParanoid 2.0.9 (Cosentino, Sriswasdi, and Iwasaki 2024), ProteinOrtho 6.3.6 (Klemm, Stadler, and Lechner 2023), FastOMA 0.3.5 (Majidian et al. 2025) and OrthoMCL 1.4 (Li, Stoeckert, and Roos 2003). These method references do not replace run-specific executable provenance. OrthoFinder’s sequence-only MCL checkpoint was a diagnostic output, not a separately finalized phylogenetic analysis. FastOMA used a supplied OrthoFinder species tree. QfO inputs included native ortholog pairs, native post-clustering relations or group-derived cross-species pairs as appropriate; these are not interchangeable output semantics. The generated comparison reports each conversion and prediction count.

OrthoBench measures curated group recovery (David M. Emms and Kelly 2020). QfO (Quest for Orthologs consortium et al. 2016; Altenhoff et al. 2020) reports GO and EC similarity, VGNC, SwissTrees and TreeFam-A F1, and FAS separately. Their arithmetic mean is an unchanged project-defined descriptive secondary summary, not an official QfO F1 or overall accuracy endpoint. It combines distinct benchmark quantities and is not used for significance, headline superiority claims or selection of a winning method. All individual endpoints and coverage remain visible. Three Kingdoms is supplementary BUSCO-reference recovery (Manni et al. 2021), not genome-wide orthology truth. QfO and OrthoBench were repeatedly inspected during development. A retained family-evidence inventory scanned 1,785 immutable JSON result blobs at Git revision `f145fff1` and 114 separately hash-frozen local-only reports, including historical parameter sweeps. The exact 70 OrthoBench/18 retained SwissTrees identities were canonicalized; score blocks, named/list mentions and unresolved ordinal/synthetic labels remained separate. Every block retained its file, JSON pointer, count-vector digest and recorded partition. Reports were not counted as independent native experiments or evidence of causal tuning. Inventory protocol. The YGOB protocol, using the curated homology/synteny resource (Byrne and Wolfe 2005), froze evaluation before test-score inspection; overlap assessment limits its interpretation to novel-taxon transfer rather than family-disjoint validation.

Supported paired uncertainty analyses resample declared reference units and recompute the benchmark statistic within replicates. They retain prespecified multiplicity adjustments, negative findings and unavailable contrasts. Curated families are not automatically independent: shared evolutionary history and predictions joining families can violate exchangeability. Dependent protein pairs are not treated as independent observations to obtain narrower intervals.

Native QfO bars are reported separately from those paired analyses. For VGNC, recall is TP/23934 and precision is TP/(TP+FP); the retained scorer stores 1.96*sqrt(p*(1-p)/n) in the generic stderr field, with the respective denominator n. These approximate binomial half-widths accompany precision and recall, not the derived F1. The incomplete negative reference and conservative false-positive rules remain limitations. For GO/EC, the retained Darwin Stat implementation defines StdErr as InverseStudent_t(0.95,n-1)*sqrt(sample_variance/n), a native 95% mean half-width, not one SEM; n counts eligible scored pairs. Means and half-widths were computed before six-decimal raw serialization. The existing arithmetic and runtime audit records this convention and its rounded-data checks. The checked corrected execution uses the same pinned Darwin image and benchmark-script hashes. GO uses eligible experimental annotation evidence; EC retains the limitations of enzyme-only coverage and possible sequence-based annotation transfer. These native bars do not account for shared proteins/families or supply paired method-difference uncertainty. Separate marginal-bar overlap is not a significance test, and bars are not propagated into VGNC F1 or the six-metric mean.

Pooled-reference TreeFam-A scores are descriptive supporting evidence because the original family trees and mapping have not been recovered. Existing FAS ranges are reported only for their stated conditional expected-statistic target and assumptions, not as biological intervals around observed scores. Missing stronger intervals for these endpoints are reporting limitations, not publication completion prerequisites. Inferential claims instead use the supported paired analyses with their development-exposure and generalization limits. This reporting scope was approved after the results; it is not represented as a prospectively specified new experiment.

The prespecified QfO parameter neighborhood retained seven arms: the frozen control and six one-at-a-time changes. CPM resolutions were 0.08/0.12 versus 0.1, candidate minimum normalization 0.024/0.036 versus 0.03, and minimum margin 1.2/1.8 versus 1.5. SwissTrees F1 was recomputed from macro precision and recall in each of 100,000 shared family-bootstrap draws, not averaged across family F1 values. The seed was 20260925; Bonferroni adjustment retained all 18 planned endpoints across six contrasts and three metrics. This is development-exposed local sensitivity, not independent method selection. Frozen parameter protocol.

Two evolutionary simulation panels used Zombi and Pyvolve (Davín et al. 2019; Spielman and Wilke 2015), ten seeds and seven conditions each: baseline, divergence, duplication/loss turnover, their combination, missing proteins, uneven sampling and a taxon-count control. Event-derived cross-species truth was not replaced by ancestral-family membership. Fixed 300-residue sequences and heterogeneous ancestral-family lengths of 100-500 residues were analyzed separately; descendants retain family length, without within-family indels. Native admission, including finite comparator graph weights, preceded scoring. Within each panel, 20,000 paired successful-seed resamples recomputed mean seed-level metrics, with a 14-contrast adjustment for primary F1. Failures were not imputed; available-case means from different seed sets are not paired effects. Simulation protocols and results.

On the 70 heterogeneous-length datasets, phylogenetic OrthoHMM and full OrthoFinder also used supplied generating trees and deterministic rooted nearest-neighbor interchanges at clade distances two and four. Generating trees are oracle diagnostics, not achievable end-to-end inference or guaranteed accuracy upper bounds. All 126 F1/precision/recall endpoints were exploratory, with 20,000 paired-seed resamples, seed 20260918 and Bonferroni adjustment. Tree-control protocol.

The subsequent fixed-candidate gene-tree diagnostic retained all 70 OrthoHMM generating-species-tree cells. Only inferred, single-ancestral-family candidates received generating gene trees, either at their generating root or rerooted by the frozen minimum-duplication/loss procedure. Candidates, species trees, reconciliation and satellite-constraint policy were fixed; native pairs had to reproduce exactly before intervention. Bypasses and mixed-ancestry candidates retained native predictions. Descriptive ten-seed

means and counts were reported without population intervals or independent-confirmation claims. The oracle changes branch lengths/support as well as topology/root, and is not an accuracy upper bound or a topology-only causal control. Gene-tree diagnostic protocol.

A retained upstream trace followed all 163,527 true pairs through significant initial HMM hits, direct final graph edges, undirected connectivity, final candidates and native pairs. The saved graph precedes refinement/expansion; it is neither initial search evidence nor ortholog output. Missing significant hits do not distinguish prefilter rejection, score filtering, caps or ranking. Gene-to-seed membership inside merged candidates was not reconstructed from incomplete sidecars. Independent igraph/SciPy component and native-row checks covered all 70 cells. Upstream trace protocol.

Finally, a complete post hoc cohort selected every generating-root candidate with within-candidate FP/FN after screening all 10,125 candidates. All cross-species pairs in those candidates, including correctly classified pairs, were traced against original simulator event tables and reconciled XML. Induced ancestor clades, species overlap, raw pair calls, root groups and native satellite-constraint filtering had to reproduce original truth and oracle counts. An independent XML/Biopython reader recalculated fixed-rule and constraint decisions without importing the trace worker. No defaults or endpoints changed. Residual trace protocol.

A post hoc reproducibility diagnostic compared accepted candidate-merge traces from the original OrthoBench satellite arm and three already completed native repeats by round and named source/target memberships. All common cluster indices and forward/reverse hit counts were checked. A controlled 19-gene fixture used the unchanged frozen candidate engine, four attachments per anchor per round and two rounds, varying only initial cluster order or one satellite's scores by one floating-point unit. Rejected candidates and the origin of historical numerical differences were not reconstructed. Candidate-cap diagnostic.

Controlled Fragment Observation Test

One prospectively frozen observation condition reuses the ten variable-length baseline histories (seeds 20261101–20261110), not new biological histories. Within each dataset, SHA256 of UTF-8 `seed:condition:gene_id` ranks all gene IDs; the first floor(N/5), with ID tie-breaking, retain a centered floor(3L/5) substring starting at floor((L-floor(3L/5))/2). Identifiers, species ownership, unselected sequences and the byte-exact parent orthology truth stay unchanged. Independent readback verified all 8,200 genes and 1,636 truncations. This is controlled synthetic truncation, not natural fragment annotation or a new independent generalization set. The simulator does not add within-family indels or validated domain architectures in this condition.

All 30 fresh native runs completed: high-sensitivity OrthoHMM (HS), satellite_v2 phylogenetic OrthoHMM (PHY), and full OrthoFinder 3.1.5 (OF), sequentially for each seed. The OF sequence-only MCL checkpoint is an additional diagnostic output, not a separately rerun final sequence-only workflow or timing estimate. Forty retained baseline outcomes are reused without rerunning controls. Settings are frozen except input/output/metrics paths. The OrthoHMM CPU budget is four with four threads per worker; actual metadata reports one search worker and four total search threads, not four simultaneous workers. The sequential job received 16 CPUs and 16 GiB on the shared Threadripper. The prospective runtime amendment records 72 full-inventory differences, while eleven required scientific dependency versions and the full OF inventory remain unchanged. This is not proof of historical OrthoHMM output equivalence.

Micro cross-species pair F1, precision, recall and pair-endpoint coverage are calculated per seed; displayed means are arithmetic means across the ten seeds, not pooled-pair F1 or a harmonic mean of mean precision and recall. Coverage is the fraction of all input genes present in at least one predicted cross-species pair, not recall. Zero/one/two truncated-endpoint strata use the same prospective flags in both arms and include inter-origin-family false positives. Their integer TP/FP/FN counts partition each whole-dataset count.

The fixed 15-endpoint comparison family comprises fragment-minus-baseline for HS, PHY and OF plus fragment HS-minus-OF and PHY-minus-OF, each for F1, precision and recall. Whole seeds, not dependent gene pairs, are resampled 20,000 times with PCG64 seed 20261011 and linear quantiles. Nominal intervals use .025/.975 and fixed-15 Bonferroni intervals use .05/30 and 1-.05/30. These are conditional percentile approximations with only ten seed units; extreme-tail coverage is not guaranteed. All planned seed lists are complete. No checkpoint comparison interval or subgroup interval is added after seeing results. Frozen protocol, runtime amendment, executed validation.

Retained Fragment Representative Protocol

Representative selection is retrospective: aggregate results had already been inspected. Four methods, three truncated-endpoint strata and seven categories define 84 bins. Categories are fragment FN/FP, new FN/FP relative to baseline, recovered FN, removed FP and retained TP controls. Within each nonempty bin, the minimum SHA256 of UTF-8 `method:category:seed:gene_a:gene_b`, with lexical tie-breaking, selects one eligible pair-record across the ten retained seeds. Thirty-one bins are empty, without replacement. Fifty-three selected records are unique in this realization; the protocol permits category overlap and does not make these observations independent or representative of prevalence.

Read-only tracing retains both arms for every selected case. Significant-hit NumPy checkpoints provide exact directed rows/scores; the weighted graph and groups provide direct edges, connectivity and seed membership. PHY additionally uses saved candidate families, observed reconciliation-node ancestors, logged merge constraints and final RootHOGs. Positive-paralogy excludes observed duplications but permits mapping-only uncertain calls; membership filtering is active only when saved constraints are nonempty. Final RootHOG separation alone is not proof of native pair exclusion. Unambiguous families without tree inference are separately marked as bypasses. OF is traced only through its MCL checkpoint and native pair output, not a matched search or event adapter.

Independent native-file readers reproduce all 106 observations in 44 contexts, including graph connectivity by BFS and bottom-up root/constraint reconstruction. The initial readback refused the TSV's null serialization after native checks; a separate tested correction accepts the original `NA` cells, not empty or false fields, while preserving the failed reader and original report/table. No inference, scoring, bootstrap or tuning is repeated.

Fresh Native Ablation Protocol

A separate frozen execution panel comprises six fresh OrthoBench identities and seven fresh corrected-QfO identities. P denotes downstream profile expansion, C candidate-family expansion and R reconciliation; initial sensitive HMM search is on in every cell. The retained P1/C0/R0 configuration is a separately reused historical baseline, not an additional fresh run. Native inference uses the frozen method with fresh output directories and does not reuse phylogenetic checkpoints. These identities are distinct from the completed 27-attempt scaling panel. Terminal scheduler, scientific-output and resource-accounting reviews remain separate gates; recovery of valid scientific outputs after a wrapper or measurement failure does not repair the original timing result. Frozen native plan.

Fresh RefOG or SwissTrees count vectors and metadata must match retained family records exactly before their original paired intervals can be reused. OrthoBench retains all 12 planned simple effects and the 36-endpoint adjustment, with 20,000 original draws (seed 20260918). QfO retains all 14 planned contrasts and the 42-endpoint adjustment, with 100,000 original draws (seed 20260922). No new bootstrap draws, independent confirmation, family-disjoint validation or total-HMM-versus-non-HMM comparison follows from this binding. Reconciliation also changes the QfO output from group-clique relations to inferred ortholog pairs.

A read-only functional diagnostic binds six native raw GO/EC/FAS tables through their original scientific admissions. It preserves original means and denominators while comparing shared and exclusive scored pairs. Independent CSV/Decimal parsing and SQLite joins check the actual tables without importing the primary comparison parsers. These are composition and dependence diagnostics, not confidence intervals, corrected endpoints or another annotation-scoring run. Diagnostic protocol.

A separately reviewed candidate diagnostic replays all accepted group unions and joins changed scored pairs through the original protein map and scoped original database rows. An independent standard-library graph traversal checks whole partitions and all pair paths without importing the primary union implementation. Group directness and transitivity describe software connectivity, not calibrated homology confidence or causal biology. The initial failed join remains retained, not rewritten as successful. Identity/path protocol.

Retrospective Native-Cell Comparator Protocol

A separately frozen retrospective analysis compares all eight planned native cells with full OrthoFinder 3.1.5 and its sequence-only MCL checkpoint, for SwissTrees F1, precision and recall. Four cells are admitted, leaving 24 of the 48 endpoints estimable; missing endpoints remain in the correction. The 18 common families contain 563 disjoint represented proteins and 10,765 reference relations. Raw counts use the official halving/unit-prior conversion, equivalent to PPV=(TP+2)/(TP+FP+4) and TPR= (TP+2)/(TP+FN+4). Every replicate recomputes harmonic F1 from macro-family precision and recall, not mean family F1.

The calculation uses 100,000 shared multinomial family draws, PCG64 seed 20260920, nominal percentile quantiles 0.025/0.975 and separate 48-endpoint Bonferroni quantiles 0.05/96 and 1-0.05/96 with linear interpolation. This is new computation, not interval transplantation from the earlier 24-endpoint comparator or 42-endpoint internal factorial analysis. Both remain unchanged. Independent literal-count readback checked statistics, all 24 intervals, family outcomes and all 48 output rows; deterministic replay is verification, not another dataset. Development exposure, family exchangeability, merged- prediction dependence and finite tail resolution limit this conditional analysis; it is not selection-adjusted or independent confirmation. Frozen protocol, actual execution/readback.

Four-Cell Profile Fixed-Stratum Extension

The retrospective extension projects admitted P1/C0/R1 counts into the same 23 fixed bins used for the three profile-off cells: 11 sequence, five domain, four reference-duplication and three model-distance bins. It adds only the conditional P1/C0/R1-minus-P0/C0/R1 contrast. Original memberships, cutoffs, integer counts, halving/unit-prior conversion and macro-family harmonic F1 are unchanged. Empty bins remain NULL/NA. All 69 inherited scores and 46 inherited differences reproduce within absolute 1e-12; the independent rational readback checks 72 family, 92 score and 69 contrast records.

The original export failed on two retained vocabularies for native pairs. A separately tested compatibility adapter normalizes only a copied metadata view and records all three original labels; retained evidence is not edited. No inference, scoring, feature extraction, bootstrap, subgroup interval, cutoff tuning, default selection or new admission occurs. Initial HMM search stays on. The older three-cell descriptions and figure below retain their original scope; this extension supplies the previously absent profile contrast. Frozen protocol, compatibility amendment, executed independent reader.

Fixed-Stratum SwissTrees Error Analyses

A complete three-cell diagnostic projects already-admitted family counts into the previously frozen sequence, Pfam and reference-duplication descriptors. All 18 families, 563 canonical members and all 54 count rows, including TN, are retained. Original prior-adjusted macro-family precision/recall followed by harmonic F1 is preserved; no raw scorer, bootstrap or original tree traversal is rerun. All 60 score and 40 conditional-difference records are retained, including empty bins as NA. Overlapping bins and repeated all-family records are not independent tests. Fixed-bin protocol, complete independent readback.

A separately prespecified feature procedure reused every retained reference alignment, without realignment, masking or filtering taxa/sites. Installed IQ-TREE 3.0.1 (Wong et al. 2026) inferred one tree per family using fixed WAG+G4, seed20261007, one thread and identical-tip retention, without ModelFinder, supplied/species trees, support resampling or a clock. The primary family descriptor is the median of all unordered tip-pair patristic distances in model-estimated expected amino-acid substitutions per site; paralog, within-species and zero-distance pairs remain. A median-of-family-medians cutoff assigns <=median to the lower bin and >median to the higher bin, with no outcome-selected alternatives. Every selected family must succeed before any cutoff/projection. The feature stage never reads prediction outcomes. Independent edge-split summation checks all 10,765 distances using shared Bio.Phylo parsing, not a second inference/model method. A separate exact-rational reader verifies every derived score and percentage-point contrast. Prospective model-distance protocol, complete feature readback, independent score readback.

Shared-Host Resource Measurement

The replacement resource panel uses the local x86 Threadripper host, bizon, not the historical DGX/ARM environment. The frozen panel comprises high-sensitivity OrthoHMM, satellite_v2 OrthoHMM with inferred phylogeny and full OrthoFinder 3.1.5 on nested sets of 4, 8 and 12 proteomes, with three rotated repeats per method and size (27 sequential attempts). The input counts are 73,266, 165,168 and 251,378 proteins. Scientific settings, input bytes and order remain frozen. Native tasks use the same 32-CPU affinity and 128-GiB ceiling; each non-exclusive Slurm allocation reserves 64 slots to accommodate measurement and reporting work. The selected affinity uses distinct physical cores, but allocation does not reserve the host against unrelated processes.

Under the user-authorized 3 October amendment, background analyses do not invalidate a run merely by competing for resources. Launch capacity, source and input identity, process attribution, measurement continuity and failure handling still apply. Process snapshots and pressure observations bracket the native interval and retain background CPU demand, memory pressure and I/O diagnostics. Snapshot coverage is not continuous isolation certification, and CPU pressure can exceed nominal bounds because of non-atomic accounting; pressure magnitudes are diagnostic rather than timing-eligibility thresholds.

The primary endpoints are native-command monotonic elapsed time, native task-subtree CPU-stat change including wrapper work, and native-step lifetime memory peak including the launcher. They exclude separate preparation, conversion and scoring; the peak is not pure algorithm RSS. Terminal scheduler,

runtime, environment, resource and native-output evidence is reviewed before the next identity. All attempts and failures are retained without fastest-run selection, automatic retries or estimated overhead subtraction. Median/range summaries require all three eligible repeats; observed ranges are not confidence intervals. Shared-host distortion is unknown and potentially method dependent. Historical DGX and earlier shared-host timings are not pooled with this panel.

Timing measurements were collected on a shared Threadripper while other analyses were running. Competition for CPU, memory bandwidth and I/O may have affected elapsed times, with an unknown and potentially tool-dependent impact. These are observed shared-host timings, not estimates of isolated performance. The 6 October authorization does not require the DGX or a quiet window; safe capacity and valid accounting remain launch prerequisites. Execution amendment, endpoint and continuation contract.

Results

Accuracy Depends On The Endpoint

Phylogenetic OrthoHMM achieved 74.1061% OrthoBench F1 versus 72.7365% for full OrthoFinder. In the exploratory eight-method comparison, the F1 difference was +1.370 percentage points (21-endpoint adjusted interval [-7.281, 12.166]). The precision difference was +15.705 [1.175, 30.634] and recall difference -13.151 [-26.947, -0.609]. This is a precision-recall tradeoff, not established F1 superiority. The complete panel uses 100,000 paired RefOG draws and remains conditional on family exchangeability and development exposure. All 21 contrasts. The complete interval figure shows F1, precision and recall on the same difference scale without ranking methods by their observed effects. A fresh installed full run reproduced all 59,770 OrthoHMM groups exactly; this establishes reproducibility, not independent accuracy or a controlled runtime comparison. Full-run verification. A second full run using a separately reconstructed base Python and fresh inference/reader environments also reproduced all groups and all 70 family score records. Native pair, confidence, reconciliation-event and hierarchy files were byte-identical to the earlier run. This extends same-host environment reproduction, not biological validation or complete archive restoration. Reconstructed-base verification. A third full run used restored local execution assets and separately acquired upstream inputs. Its independent scientific admission reproduced all 59,770 groups, all 70 family score records and four native TSVs exactly, without reusing phylogenetic checkpoints. This validates local archive-to-results reproduction, not cross-host/OS restoration, independent accuracy or controlled timing. Restored-archive verification.

On corrected QfO, phylogenetic OrthoHMM versus full OrthoFinder scored 0.901690 versus 0.988546 on VGNC, 0.833513 versus 0.848413 on SwissTrees, and 0.614864 versus 0.791918 on TreeFam-A. OrthoHMM had higher GO similarity (0.490349 versus 0.469548), EC similarity (0.965650 versus 0.936130) and FAS (0.762993 versus 0.691422). These endpoints differ in reference scope and eligibility; they do not form a single accuracy ranking. All eight methods.

The supplementary native-bar tables below retain all eight methods' VGNC precision/recall and GO/EC means and eligible scored-pair counts. Their conventional half-widths are descriptive benchmark-standard reporting, not paired significance or evidence of unseen-dataset superiority. TreeFam point differences remain descriptive supporting evidence.

An exploratory VGNC deletion diagnostic removed each of 16,844 reference blocks and its incident scored pairs in turn, across all eight methods. Every comparator retained a negative F1 difference against full OrthoFinder for every deletion. For phylogenetic OrthoHMM the range was [-8.7353, -8.6729]

percentage points. These are sensitivity ranges, not confidence intervals: dependent deletions change the fixed scored table, and single-block sign stability does not establish population or joint-deletion robustness. Native inference and eligibility were not rerun.

For corrected SwissTrees, phylogenetic-minus-full-OrthoFinder F1 was -1.4900 percentage points, with a multiplicity-adjusted interval of [-8.7078, 7.2090]. Phylogenetic-minus-high-sensitivity OrthoHMM was +14.8015 points [4.9791, 26.6015]. These conditional estimates use 18 development-exposed families and a 24-endpoint adjustment; inclusion of zero is not equivalence. Candidate expansion also changes between OrthoHMM configurations, so their contrast does not isolate reconciliation. Protocol and contrast evidence.

All 56 method-pair GO/EC comparisons had identical six-decimal scores on shared scored pairs. Aggregate differences arose from different eligible pair sets and their denominators. Restricting evaluation to intersections would change the endpoint rather than resolve uncertainty. Historical raw hashes were verified for all 16 input tables. Scored-pair audit. The complete pair-composition figure displays the shared-denominator and exclusive-pair terms for all seven comparators against full OrthoFinder. Phylogenetic OrthoHMM's shared pairs account for 90.12% of its GO scored set but 46.40% of OrthoFinder's, and 86.59% versus 58.00% for EC. The original rounded-mean differences, +2.080130 and +2.952029 score points, are the sums of much larger opposing terms. These fractions describe eligible scored pairs, not proteome-wide coverage; the decomposition is arithmetic, not causal attribution or paired uncertainty.

Component Evidence Is Bounded

The corrected QfO factorial supported candidate-expansion-by-reconciliation interactions on SwissTrees F1, while all four profile-refinement F1 intervals included zero. Reconciliation increased precision and reduced recall. These results support conditional component effects, not proof that profile expansion universally improves orthology inference. Complete factorial.

In 35 matched-recall simulation datasets spanning seven conditions and five seeds, HMM-derived search evidence yielded mean downstream graph F1 of 83.6097%, versus 80.5180% for DIAMOND (Buchfink, Reuter, and Drost 2021). The adjusted paired seed-block interval for the +3.0917-point difference was [1.6782, 4.5313]. Profile expansion and phylogeny were off. This is a development-exposed fixed-graph comparison, not a comparison against OrthoFinder or a matched-effort result. Score rankings and hit identities differ, so the design does not isolate a causal mechanism. Matched-recall control.

Fresh Native OrthoBench Effects

All six fresh OrthoBench cells have admitted scientific scores over 70 RefOGs; 420 family records match the retained factorial exactly. Partition agreement is not asserted for every non-reference gene. P0/C0/R0 retains failed wrapper 22427 with scientific recovery, not successful timing. The two omitted native cells and their contrasts remain unavailable rather than imputed.

Native Cell	Group-Recovery F1 (%)	Precision (%)	Recall (%)
P0/C0/R0	69.7634	78.8686	62.5429
P0/C0/R1	72.7050	87.4948	62.1922
P0/C1/R0	66.6395	64.4883	68.9391
P0/C1/R1	73.4023	81.4862	66.7776

Native Cell	Group-Recovery F1 (%)	Precision (%)	Recall (%)
P1/C0/R1	73.3114	87.4586	63.1038
P1/C1/R0	67.2074	64.7001	69.9169

Native point estimates, conditional interval binding, independent family-record readback.

Six of 12 planned simple effects have matching native records. With C0/R1 fixed, profile expansion changes F1 by +0.6064 percentage points (adjusted interval [-1.2114, 4.5210]; family wins/ties/losses 4/61/5). With C1/R0 fixed, its effect is +0.5680 [-1.1132, 4.2906] (6/59/5). Both intervals include zero. These are downstream-profile effects with initial HMM search retained, not the total HMM contribution or proof of general benefit.

With P0 fixed, reconciliation changes F1 by +2.9416 [0.2542, 8.1443] at C0 and +6.7628 [1.3762, 14.6116] at C1. Candidate expansion changes F1 by -3.1239 [-10.6588, 3.2751] at R0 and +0.6973 [-2.4590, 4.8209] at R1, while increasing recall and reducing precision in both cases. These conditional development-exposed group-recovery contrasts retain family-exchangeability and percentile-coverage limitations. They neither complete a fresh eight-cell factorial nor establish superiority over full OrthoFinder. Group co-membership is not resolved pairwise-ortholog truth. Interpretation and exact endpoints.

Fresh Native QfO Trade-Off And Pair Composition

Four of seven fresh corrected-QfO cells have admitted accuracy; three remain unavailable in this dated snapshot. Initial sensitive HMM search stays on in every cell. P denotes downstream profile refinement, C candidate expansion and R phylogenetic reconciliation. This is not the selected-default all-tool comparison above or a total-HMM-off control. The recovered P0/C0/R1 result retains failed timing, null resources and timing ineligibility. Accuracy admission does not repair that failure.

Endpoint	Statistic	P0/C0/R0	P0/C0/R1	P0/C1/R0	P1/C0/R1
VGNC	F1	0.66683353	0.89818458	0.64744548	0.89698058
SwissTrees	F1	0.68918396	0.78957434	0.68570950	0.78637219
TreeFam-A	F1	0.60540356	0.60250783	0.63366391	0.60146776
GO	Similarity	0.47211905	0.49025964	0.47053999	0.49040899
EC	Similarity	0.93211406	0.96770222	0.92062409	0.96758244
FAS	Sample mean	0.77494473	0.78500682	0.75030314	0.78501436

Admitted native endpoint snapshot. The project-defined secondary means are 0.69009982, 0.75553924, 0.68471435, 0.75463772, respectively, not official QfO F1 or a single accuracy ranking. Prediction coverage uses all 984,137 input accessions, not the number of assessed reference pairs. Mapping loss is zero in these conversions.

Cell	Submitted pairs	Input-accession coverage	Output semantics
P0/C0/R0	9,009,082	55.5258%	Group-clique pairs
P0/C0/R1	5,113,820	55.1107%	Inferred ortholog pairs

Cell	Submitted pairs	Input-accession coverage	Output semantics
P0/C1/R0	11,734,021	59.5049%	Group-clique pairs
P1/C0/R1	5,115,410	55.1078%	Inferred ortholog pairs

Three of 14 planned contrasts have exactly matched native SwissTrees family records; the other 11 remain unavailable. The original 100,000 paired-family draws (seed 20260922) and all 42 adjusted endpoints are reused, with no new draws. Effects and adjusted intervals below are percentage points; wins/ties/ losses count the 18 families, not independent gene pairs.

Contrast	Endpoint	Difference (pp)	Adjusted interval (pp)	Wins/ties/losses
C at P0 R0	F1	-0.3474	[-5.8168, 3.3091]	5/8/5
C at P0 R0	Precision	-3.5235	[-10.2296, 0.3740]	4/8/6
C at P0 R0	Recall	+4.3752	[0.9401, 8.6887]	9/9/0
R at P0 C0	F1	+10.0390	[-4.6418, 24.4199]	14/1/3
R at P0 C0	Precision	+30.5212	[14.3252, 46.5473]	16/1/1
R at P0 C0	Recall	-6.5334	[-18.0413, -0.0436]	0/10/8
P at C0 R1	F1	-0.3202	[-2.1129, 0.5132]	1/16/1
P at C0 R1	Precision	-0.0101	[-1.3228, 1.2725]	1/16/1
P at C0 R1	Recall	-0.4630	[-2.3148, 0.0000]	0/17/1

All three adjusted F1 intervals include zero. Candidate expansion raises recall while lowering precision; reconciliation shifts that trade-off in the opposite direction. Profile refinement at C0/R1 has a slightly lower F1 point estimate, with one family improving, 16 tying and one worsening. These are conditional pipeline contrasts, not identified main effects, interactions, equivalence or independent confirmation. Count-based arithmetic differs slightly from native serialized endpoints and remains separately reported. Development exposure, family exchangeability and approximate percentile coverage limit the intervals; they do not apply to other QfO endpoints or the secondary mean. Guarded interval binding, independent arithmetic readback.

Native QfO Figure. The reviewed six-panel figure separates F1 from functional similarities and coverage, and shows the three condition-specific SwissTrees contrasts. It is a four-cell ablation, not a complete fresh factorial or a timing comparison. All 24 endpoint scores, four coverage rows and nine intervals were separately read back; both actual PNG and decoded PDF preview were visually inspected. Figure evidence.

Every changed assessed SwissTrees profile pair was traced, avoiding favorable example selection. Three CASP false positives disappeared and one GH14 true positive was lost. All four pairs shared a candidate family with a speciation pair-event before refinement, then spanned different candidate families afterward. Thus these absences already follow from candidate separation before reconciliation, not a new duplication exclusion at a shared LCA. Saved Newick and independent root-membership checks cover six families, 70 leaves and the two before-state LCAs. Species-tree byte hashes also differ. These checks explain software inclusion, not biological correctness, individual HMM-edge causality or an isolated species-tree effect. Equal family weighting and different precision/recall changes explain why removing three false positives while losing one true positive need not improve macro F1. Complete localization and limits, independent saved-tree check.

The earlier three-cell table, binding and figure remain historical and unchanged. Their fixed-bin diagnostics below still describe P0 cells and do not estimate the new profile-refinement effect.

Candidate expansion recovers 162 asserted VGNC TPs and adds 2,133 scored FPs while retaining baseline TPs/FPs. Across 42,080 union rows, precision changes -2.904231 percentage points, recall +0.676861 and F1 -1.938806. All added FPs cross fixed reference overlap blocks. Those blocks are not validated independent uncertainty units; an unscored baseline pair is not a true negative. This explains the score arithmetic, not biological orthology or causal false-homology mechanisms. VGNC decomposition, independent readback.

The complete accepted-union reconstruction preserves 984,137 genes and transforms 394,328 baseline groups into 353,638 candidate groups through 40,690 unions. The initial export failure is retained; a separately versioned follow-up proves the two scored/native accession bridges from the original protein map and DB rows, without suffix guessing, then independently localizes every one of the 2,295 changed scored pairs.

Candidate round	Group-connection path	Recovered TPs	Added FPs
0	Direct group endpoints	131	1,794
0	Transitive union	11	98
1	Direct group endpoints	19	239
1	Transitive union	1	2
Total	All localized paths	162	2,133

Most added FPs have direct group-endpoint connections, ruling out an all-transitive software-path explanation. Directness here does not prove a direct protein-pair HMM hit, causal erroneous evidence or a biological mechanism; transitivity does not prove absence of supporting evidence. Original scored identifiers and categories remain unchanged. Complete localization, full pair ledger, independent graph readback, alias proof and retained failure.

For P0/C0, every R-on GO scored pair (78,607) occurs in the R-off set (145,142), and every R-on EC scored pair (116,929) occurs in R-off (186,098). All common serialized six-decimal scores agree exactly. The 66,535 GO and 69,169 EC R-off-only pairs have lower means, 0.4506870677 and 0.8719528955. Thus, at retained precision, the higher native functional means reflect scored-pair membership and denominator composition, not changed similarities on common pairs or proof of correct ortholog selection. Intersection-only means would change the endpoint.

FAS realizes 38,205 R-off and 252,451 R-on sampled pairs, with only 1,007 shared pairs: 2.6358% of R-off and 0.3989% of R-on. All shared serialized scores agree; the original sample-mean difference remains +0.0100620873, not replaced by a shared-only difference of zero. Unseeded method-specific sample mixtures, missing scores and repeated proteins prevent this overlap from supplying an admitted paired confidence interval. Independent SQLite readback checks all 817,432 rows in the six bound tables. Neither membership arithmetic nor native pair-IID standard errors resolve other-endpoint comparison uncertainty. Functional composition result, independent SQL readback. No new score, default, independent-validation claim or timing repair accompanies this diagnostic. Stronger biological paired uncertainty for GO/EC/FAS, VGNC, TreeFam-A and the secondary mean remains unavailable; the approved descriptive and conditional reporting scope does not require it for publication completion.

Conditional Sampling In Retained Eight-Method FAS

This analysis concerns the retained September eight-method comparator snapshot, not the later four-cell fresh-native ablation above. It adds algorithmic sampling uncertainty without changing an observed score, method or default.

Condition on eligible pair populations, annotation/lookup/scoring options and the retained full precomputed mean mu_P. Assume uniform without-replacement selection and independent strata within a method, with fixed per-pair numeric return status/value under normal successful execution. If G of M missing-lookup pairs return numeric values, R is hypergeometric for c requests. The expected native ratio is theta=w*mu_P+(1-w)*mu_G, w=E[k/(k+R)], not substitution of E[R]. Count-tail inversion and conditional bounded-mean concentration are projected through this mixture. Alpha=0.05 is allocated across two components for each of eight methods; their joint rectangle supplies all 28 difference ranges without assuming cross-method independence. Score-dependent omissions are allowed, with no missing-at-random or biological pair/family independence.

Retained Method	Observed Native Z	Conditional 95% Expected-Theta Range
orthohmm_high_sensitivity	0.776585042	[0.771332967, 0.780286193]
orthohmm_phylogeny_satellite_v2	0.762993312	[0.761929649, 0.763906702]
orthofinder_3_1_5_full	0.691422133	[0.688429518, 0.694915139]
orthofinder_3_1_5_sequence_only	0.561753084	[0.546499945, 0.577724179]
sonicparanoid_2_0_9	0.736680187	[0.732115268, 0.738512044]
proteinortho_6_3_6	0.813594864	[0.813565174, 0.813689528]
fastoma_0_3_5	0.654366336	[0.643583534, 0.662539506]
orthomcl_1_4	0.733752216	[0.731706501, 0.735008398]

Observed Z and expected theta are distinct: unknown G/mu_G preclude an exact theta point value. These ranges are not biological error bars centered on Z. Both OrthoHMM modes exceed full OrthoFinder on this conditional FAS target, but Proteinortho exceeds both; high sensitivity exceeds phylogenetic OrthoHMM. SonicParanoid versus OrthoMCL overlaps zero. All favorable and unfavorable contrasts are retained. This is not overall orthology superiority or a causal explanation, and earlier negative F1 comparisons remain unchanged.

The prospective finite-population validation covers 15 cells, including four literal 9000-cap cases; its context-dependent-return control is rejected. Independent 80-digit recurrence readback checks native count endpoints, weights and all ranges/projections without producer/kernel/SciPy imports. Conditional design assumptions, unestablished historical parser/database bindings and missing historical RNG/pair identities remain limitations, not certificates. No biological generalization, other-endpoint uncertainty or readiness is admitted. Frozen reporting plan, actual conditional result and limits, all 28 contrasts, actual execution and independent readback.

Native Factorial FAS Sampling Uncertainty

The four admitted newer QfO ablations have simultaneous conditional 95% ranges for expected repeated native post-attrition FAS, accounting for unknown means in both the precomputed and newly scored strata. Initial HMM search remains on; P is profile refinement, C candidate expansion, R reconciliation. These are not the earlier September eight-tool/default comparison. Observed Z is retained separately from the expected-ratio range; unknown numeric-return population size and stratum means preclude an exact expected-ratio point.

Cell	Observed Z	Conditional Expected FAS Range	Precomputed Draws	Missing Draws	Numeric Missing Returns
P0/C0/R0	0.774944731	[0.762713985, 0.787179807]	29216	9000	8989
P0/C0/R1	0.785006818	[0.780906054, 0.789109619]	243473	9000	8978
P0/C1/R0	0.750303136	[0.736973521, 0.763638295]	23962	9000	8979
P1/C0/R1	0.785014356	[0.780955178, 0.789075399]	248211	9000	8971

All six differences use the same joint rectangle. P0/C0/R1 minus P0/C0/R0 is [-0.006273752, 0.026395633]; the approximately 0.010062 observed increase does not establish a nonzero expected-FAS effect under these bounds. P1/C0/R1 minus P0/C0/R1 is [-0.008154440, 0.008169344], not proof of equivalence or of an additional profile-refinement benefit. P0/C0/R1 minus P0/C1/R0 is [0.017267759, 0.052136098], and P1/C0/R1 minus P0/C1/R0 is [0.017316883, 0.052101878]. These compare configurations differing in more than one component, so they do not isolate a causal reconciliation effect. P0/C0/R0 minus P0/C1/R0 is [-0.000924310, 0.050206286], also including zero. All orientations, observed differences and full-precision values are retained in the linked result, including unfavorable and unresolved comparisons.

The derivation conditions on fixed pair populations, lookup/annotations/options, and context-invariant numeric-return status and values. Within each cell, uniform without-replacement selections in the two strata are assumed; actual historical RNG states remain unavailable. A count-tail interval and two bounded sample-mean intervals receive error 0.05/12 per component across four cells. Eight-corner projection retains the random native denominator rather than substituting a plug-in return count. No biological gene-pair IID, independent families, cross-cell independence or missing-at-random condition is assumed. The prior finite native context experiment is supportive only for its tested pairs, not proof of all historical batch/worker outcomes.

Successful native task identity, admitted execution/raw bindings and logged counts are checked. Precomputed-first raw ordering in the pinned native writer recovers exact sampled stratum means and reproduces the rounded logged means and full-precision aggregate. The four task logs were newly hash-pinned and were not historically hash-bound in the original admission records. No full precomputed population mean, database join or omitted-pair reconstruction is substituted. The native pair-IID SEM is not used as a biological interval.

The tested source/protocol was pushed as 88ddb696 before first application. All 84 focused/adjacent tests passed. Actual reporting completed four ranges and six contrasts without numerical failure. An independent 80-digit relative probability recurrence checks all 16 count endpoint/neighbor conditions, both raw stratum means, mean bounds, weights, corner ranges and six contrasts; maximum numerical disagreement was 7.044019369886139e-16. This is numerical and conditional-design validation, not biological confirmation. Retained scores, negative F1 comparisons, three unavailable factorial cells and P0/C0/R1's null/ineligible failed timing are unchanged. Biological generalization and other-endpoint uncertainty remain unresolved; they are documented reporting limitations rather than new completion gates. These results alone do not prove publication readiness. Earlier manuscript statements of unavailable FAS uncertainty precede this conditional extension and do not describe its biological scope.

Frozen derivation and reporting plan, all four ranges and six contrasts, generated table, actual execution and independent readback.

Retrospective Native-Cell Comparator Sensitivity

Native minus OrthoFinder SwissTrees F1 differences and adjusted intervals below use raw 0-to-1 units. Family outcomes are descriptive, not independent pair counts. All planned missing comparisons remain unavailable.

Native Cell	Comparator	F1 Difference	48-Endpoint Adjusted Interval	Family Wins/Ties/Losses
P0/C0/R0	Full OrthoFinder	-0.159229	[-0.272661, -0.007023]	1/0/17
P0/C0/R1	Full OrthoFinder	-0.058839	[-0.164040, 0.018543]	3/4/11
P0/C1/R0	Full OrthoFinder	-0.162704	[-0.298476, 0.013904]	1/0/17
P1/C0/R1	Full OrthoFinder	-0.062041	[-0.168638, 0.018129]	3/4/11

Native Cell	Comparator	F1 Difference	48-Endpoint Adjusted Interval	Family Wins/Ties/Losses
P0/C0/R0	Sequence-Only OrthoFinder	-0.001331	[-0.094152, 0.098224]	7/5/6
P0/C0/R1	Sequence-Only OrthoFinder	+0.099060	[-0.117354, 0.302180]	12/0/6
P0/C1/R0	Sequence-Only OrthoFinder	-0.004805	[-0.081246, 0.067536]	5/8/5
P1/C0/R1	Sequence-Only OrthoFinder	+0.095857	[-0.120221, 0.300133]	12/0/6

All four F1 point estimates are below full OrthoFinder. Only P0/C0/R0 has an adjusted F1 interval excluding zero in that direction. Both unreconciled cells have lower precision with adjusted intervals below zero. Reconciled precision differences versus full OrthoFinder have intervals including zero. Relative to its sequence-only checkpoint, P0/C0/R1 and P1/C0/R1 have positive precision intervals and negative recall intervals; their positive F1 point differences have intervals including zero. P0/C0/R0 also has lower recall than this checkpoint. Eight of 24 estimated adjusted intervals exclude zero, conditionally within this separate family. This is not a whole-study significance count, equivalence test or general superiority finding. No default selection or initial-HMM causal effect follows.

Native Comparator Figure. The six-panel figure shows F1, precision and recall against full OrthoFinder (top) and its sequence-only checkpoint (bottom). Dots: point differences. Thick lines: nominal 95% intervals. Thin lines: 48-endpoint adjusted intervals. The figure alone uses percentage points; the table/TSV retain raw units. All eight cells appear in both rows; unavailable points are not zeros. Complete 48-row table, full-precision machine result, claim-to-evidence addendum.

P1/C0/R0 is absent from this supplied fresh-cell snapshot, not globally missing: the historical high-sensitivity method remains separate. The other absent cells failed before inference, during assessment or during native inference, respectively; no score is imputed. The recovered P0/C0/R1 timing remains failed/ineligible. R changes pair-output semantics as well as reconciliation; initial HMM search remains on in every cell. These intervals do not cover other QfO endpoints, the secondary mean, resource timings or missing outcomes. The terminal failure section below retains its earlier scope; its no-new-draw statement concerns that failure reporting update, not this separately specified retrospective calculation.

The terminal resource consolidation joins all 13 observed ablation attempts with explicit accuracy/failure and resource-scope flags. It does not repair failed measurements or establish isolated performance. Original shared-host contention limits remain.

Terminal Native QfO Outcomes

The later terminal reviews do not add any admitted scores: four of seven fresh cells retain admitted accuracy and three retain missing scores. The preceding four-cell endpoint table, figure, intervals and localization remain unchanged. This update is failure reporting, not a complete factorial, independent confirmation or superiority over OrthoFinder.

Cell	Accuracy status	Six-metric mean
p0_c0_r0	supplied_native_admission	0.69009982

Cell	Accuracy status	Six-metric mean
p0_c0_r1	supplied_recovered_scientific_admission	0.75553924
p0_c1_r0	supplied_native_admission	0.68471435
p0_c1_r1	no_supplied_native_admission	Unavailable
p1_c0_r1	supplied_allocated_native_admission	0.75463772
p1_c1_r0	scoring_OUT_OF_MEMORY_no_admission	Unavailable
p1_c1_r1	native_SIGSEGV_no_admission	Unavailable

Native11 (P1/C1/R0) completed inference and group-clique conversion, retaining 11,755,521 pairs and relation coverage 585,180/984,137. Its assessment ended OUT_OF_MEMORY at 8 CPU slots/32 GiB; FAS exited137. Five other endpoint tasks completed but supply no admitted score or mean. Coverage is not accuracy; its exact scoring peak memory remains unknown.

Native12 (P1/C1/R1; job24036) ended FAILED1:0. The observer native-step wrapper completed0:0, but the actual native command exited-11 (SIGSEGV). The final native receipt remained running and final metrics/orthogroup outputs were absent. Retained downstream phylogeny files are partial evidence, not successful inference or admitted pair predictions. Conversion and scoring were not run. No stack trace was captured; the root cause and crash location are unknown. The last buffered progress message does not identify the crashing operation. No automatic inference retry was performed.

Reviewed failed-command observations: 64526.103655 wall seconds, 1739733.457142 native-task CPU seconds and 20492017664 bytes native-step lifetime peak memory, with the documented wrapper/launcher scopes. These describe a failed attempt, not successful inference or scoring cost.

Shared Threadripper measurements while other analyses were running. CPU, memory-bandwidth and I/O contention may affect elapsed times by an unknown, potentially tool-dependent amount. These are observed shared-host timings, not estimates of isolated performance.

Missing outcomes remain null, not zero. The three supported SwissTrees contrasts and all 42 planned adjusted endpoints retain their original scope; no new bootstrap draws, endpoints, functional similarity claims or default selection follow from this update. Prepared success-only five-cell reporting was not executed. This revision remains not submission-ready.

Evidence: - snapshot (SHA256 `7aab1cbd31fb650df42e6e80a14ee0167b35a810f08202768c89c514755211a2`) - native11 (SHA256 `4f1a5c536af8ce992d46866374b3ab44ee3158077316d69ba2e9f5fb34cee203`) - review (SHA256 `a55a6da5ffb8dfbb362883937c915b161796a9ddd7275f4ac48a1c4d049e5c4a`)

Profile Refinement Across All Fixed Bins

All 23 conditional profile-bin differences are reported below in percentage points. Overlapping and repeated all-family bins are audit rows, not independent findings. NA denotes an empty bin, not zero. No subgroup intervals were computed.

Suite	Bin	Families	F1 Difference	PPV Difference	TPR Difference
sequence	all	18	-0.320	-0.010	-0.463

Suite	Bin	Families	F1 Difference	PPV Difference	TPR Difference
sequence	concentrated	0	NA	NA	NA
sequence	explicit_fragment	0	NA	NA	NA
sequence	higher_entropy	9	-0.763	-0.529	-0.926
sequence	lower_entropy	9	+0.133	+0.509	+0.000
sequence	missing	0	NA	NA	NA
sequence	missing_entropy	0	NA	NA	NA
sequence	no_explicit_fragment	18	-0.320	-0.010	-0.463
sequence	not_concentrated	18	-0.320	-0.010	-0.463
sequence	not_short_relative	11	-0.499	-0.016	-0.758
sequence	short_relative	7	+0.000	+0.000	+0.000
domain	all	18	-0.320	-0.010	-0.463
domain	median_pfam_types_at_least_two	6	+0.000	+0.000	+0.000
domain	median_pfam_types_below_two	12	-0.496	-0.015	-0.694
domain	repeated_type_fraction_at_least_quarter	3	+0.000	+0.000	+0.000
domain	repeated_type_fraction_below_quarter	15	-0.378	-0.012	-0.556
duplication	all	18	-0.320	-0.010	-0.463
duplication	lower_duplication_fraction	9	+0.000	+0.000	+0.000
duplication	missing_duplication_fraction	0	NA	NA	NA
duplication	upper_duplication_fraction	9	-0.680	-0.020	-0.926
model_distance	all	18	-0.320	-0.010	-0.463
model_distance	higher_than_median	9	+0.160	+0.509	+0.000
model_distance	lower_or_equal_median	9	-0.794	-0.529	-0.926

Overall F1 changes by -0.320215 percentage points. Lower-entropy families have +0.132851 points, while higher-entropy families have -0.762755. Above-median model-distance families have +0.160465, versus -0.794278 at or below the fixed median. Each entropy/distance bin has nine families.

These patterns reflect the same retained changed families rather than many independent successes. Profile-minus-reference integer count changes are:

Family	TP	FP	FN	TN
CASP	+0	-3	+0	+3
GH14	-1	+0	+1	+0

The other 16 families have unchanged counts. CASP contributes the precision gain in the lower-entropy/higher-distance bins; GH14 contributes the recall loss in the complementary bins. Both occur in the not-short-relative, lower-Pfam-type, lower-repeat-fraction and upper-reference-duplication bins. Their complementary bins are unchanged. Five empty bins support no effect. The existing complete pair localization places all four removed calls at pre-reconciliation candidate separation. It does not identify a causal HMM edge or establish correct gene/species trees.

These are development-exposed descriptive associations, not a general entropy or divergence benefit, an independent validation, or an initial-HMM effect. Existing aggregate intervals are not subgroup intervals. Fragment flags, Pfam descriptors, reference duplication fractions and model distances retain their proxy limitations. Missing factorial cells and failed/ineligible timing remain. Complete four-cell table, full-precision result, actual commands/readback.

Fixed-Stratum Errors And Estimated Sequence Divergence

Candidate expansion lowers precision and increases recall in every nonempty fixed sequence/domain/reference-duplication bin. These records overlap and include repeated all-family summaries; they are not multiple independent confirmations. The complete fixed-bin figure shows both conditional contrasts and all three metrics in all 11 nonempty, nonredundant bins (66 points). Nine excluded records are five empty bins and four redundant all-family copies, not outcome-based omissions. Complete tables retain every record. Dots are descriptive, not confidence intervals, and the two panels use explicitly different percentage-point axis ranges. Full table/readback, actual figure review.

The fixed-model distance procedure succeeded for all 18 families, without a timeout or retry. The median family-distance cutoff was approximately 2.24582463005 substitutions/site, giving nine families in each bin. The complete conditional differences were:

Distance Stratum	Families	Conditional Contrast	Delta F1 (pp)	Delta Precision (pp)	Delta Recall (pp)
All families	18	R at P0/C0	+10.039	+30.521	-6.533
All families	18	C at P0/R0	-0.347	-3.523	+4.375
Lower/equal distance	9	R at P0/C0	+11.945	+32.417	-7.950
Lower/equal distance	9	C at P0/R0	-1.245	-4.547	+5.233
Higher distance	9	R at P0/C0	+8.427	+28.625	-5.117
Higher distance	9	C at P0/R0	+0.324	-2.500	+3.517

Expansion trades precision for recall in both distance bins, with a small higher-distance F1 increase but a lower-distance and overall decrease. Reconciliation raises F1/precision and lowers recall in both. These exposed subgroup point patterns have no new intervals or significance claim; neither a causal mechanism nor a distance-dependent default follows. Initial HMM search is on and downstream profile refinement is off in all three cells. C and R contrasts are conditional, not an identified interaction. Complete model-distance result, complete scores and differences.

Synthetic Null Tails Depend On Composition

At the frozen `E < 1e-4` cutoff, width-64 scoring passed 0/1/0 of 10,000 ordinary-background pairs at lengths 50/150/400. These sparse counts do not establish rare-tail calibration. Half-background-plus-half-glutamine sampling passed 89.83%, 100% and 100%; its adjusted intervals were [88.7484%, 90.8453%] and [99.9181%, 100%] for each longer length. All 90 endpoints and their figure are retained. A hit here is only a forced pair passing the approximate significance gate, not a predicted ortholog or an observed pipeline false positive. No prefilter or biological inference ran. The strong synthetic composition is not a measured real-proteome prevalence. The audit reproduces all endpoint counts and 180 sparse reference-Python scores, not all native scores. No coefficients, thresholds or defaults were fitted or promoted.

Evolutionary Simulations Retain Failures And Tree Sensitivity

In the fixed-length stress panel, high-sensitivity and phylogenetic OrthoHMM had 70 and 64 admitted datasets out of 70. No OrthoFinder output passed its native-completion/finite-graph gate; all 14 planned comparisons are unavailable, not wins for OrthoHMM. The heterogeneous-length panel admitted 70 and 67 OrthoHMM datasets and 65 full OrthoFinder datasets. Full OrthoFinder had higher paired mean F1 than both OrthoHMM modes in all seven conditions. Adjusted F1 intervals were below zero for all seven high-sensitivity contrasts and four phylogenetic contrasts. Phylogenetic-minus-full-OrthoFinder effects were -11.7433 points for divergence (five paired seeds) and -12.2842 for combined divergence/turnover (eight). These are success-conditioned comparisons, not failure-adjusted population estimates. Species-tree inference failures for OrthoHMM and nonfinite graph weights for OrthoFinder remain explicit. Fixed stress results, heterogeneous-length results.

The complete tree-control analysis retained 560 arm outcomes: 537 scored and 23 failed. No generating-minus-inferred adjusted interval excluded zero. The stronger perturbation reduced F1 and recall with adjusted intervals below zero in four OrthoHMM conditions and two OrthoFinder conditions: 12 endpoints; no precision interval excluded zero. Retained upstream artifacts agreed in 400 comparisons, differed in two and were unavailable in 18. The differing OrthoFinder cases prevent a strict tree-only causal attribution there. These deterministic perturbations and ten planned seeds do not establish robustness to arbitrary trees or empirical posterior uncertainty. Complete tree results and figure.

Controlled Fragment Accuracy

All 40 fragment method/checkpoint outcomes were admitted across ten seeds. HS denotes high sensitivity, PHY denotes satellite_v2 with phylogeny, and OF denotes full OrthoFinder. Values below are percent; coverage is defined in the controlled observation protocol above.

Method	Baseline F1	Fragment F1	Fragment Precision	Fragment Recall	Coverage
HS	97.0772	96.9773	94.9040	99.1855	99.9635
PHY	99.4584	99.3361	99.8940	98.8060	99.9147
OF	99.9424	99.9080	99.9257	99.8906	99.9633
OF checkpoint	97.7938	97.4260	95.0881	99.9386	99.9878

Controlled Fragment Paired Differences

All 15 planned differences and interval endpoints are shown in percentage points. F-B denotes fragment minus baseline; F-F denotes the fragment method minus fragment OF. Each row has ten eligible paired seeds.

Comparison	Metric	Difference	Nominal 95% Interval	Fixed-15 Adjusted Interval	Seeds
HS F-B	f1	-0.0999	[-0.2759, +0.0000]	[-0.3878, +0.0000]	10
HS F-B	precision	-0.0188	[-0.0539, +0.0000]	[-0.0740, +0.0000]	10
HS F-B	recall	-0.1891	[-0.5196, +0.0000]	[-0.7326, +0.0000]	10
PHY F-B	f1	-0.1223	[-0.3502, +0.0377]	[-0.4835, +0.0893]	10
PHY F-B	precision	-0.0222	[-0.0836, +0.0224]	[-0.1126, +0.0376]	10
PHY F-B	recall	-0.2116	[-0.6017, +0.0815]	[-0.8371, +0.1805]	10
OF F-B	f1	-0.0344	[-0.0893, +0.0000]	[-0.1237, +0.0000]	10
OF F-B	precision	-0.0139	[-0.0347, +0.0000]	[-0.0417, +0.0000]	10
OF F-B	recall	-0.0547	[-0.1503, +0.0000]	[-0.2050, +0.0000]	10
HS-OF F-F	f1	-2.9307	[-4.6881, -1.5314]	[-5.6825, -1.0863]	10
HS-OF F-F	precision	-5.0217	[-7.2106, -3.0158]	[-8.2208, -2.1726]	10
HS-OF F-F	recall	-0.7051	[-2.2931, +0.1571]	[-3.0668, +0.2186]	10
PHY-OF F-F	f1	-0.5719	[-1.3924, -0.0404]	[-1.9815, -0.0087]	10
PHY-OF F-F	precision	-0.0316	[-0.1206, +0.0188]	[-0.1655, +0.0236]	10
PHY-OF F-F	recall	-1.0846	[-2.5981, -0.0906]	[-3.6622, -0.0244]	10

Both fragment HMM configurations have lower mean F1 than full OF in this condition; both adjusted F1 difference intervals exclude zero on the negative side. The three native fragment-minus-baseline F1 intervals include zero, which establishes neither equivalence nor absence of fragment sensitivity. No default was changed based on these outcomes.

Controlled Fragment Endpoint Counts

Counts below sum the ten seeds; they are not the mean-score or interval statistics. Endpoint count is the number of prospectively flagged genes in a pair, including the reused untruncated baseline arm.

Method	Truncated Endpoints	Baseline TP / FP / FN	Fragment TP / FP / FN
HS	0	18538 / 1017 / 116	18535 / 1017 / 119
HS	1	9169 / 521 / 66	9127 / 521 / 108
HS	2	1155 / 54 / 7	1143 / 54 / 19
PHY	0	18487 / 14 / 167	18484 / 19 / 170
PHY	1	9122 / 10 / 113	9075 / 11 / 160
PHY	2	1148 / 0 / 14	1135 / 0 / 27
OF	0	18639 / 13 / 15	18641 / 17 / 13
OF	1	9234 / 5 / 1	9216 / 5 / 19
OF	2	1162 / 0 / 0	1162 / 0 / 0
OF checkpoint	0	18654 / 853 / 0	18654 / 1006 / 0
OF checkpoint	1	9235 / 430 / 0	9217 / 493 / 18
OF checkpoint	2	1162 / 48 / 0	1162 / 54 / 0

Unflagged-pair counts can change through grouping and reconciliation even though neither endpoint was truncated. These counts alone do not localize a causal stage. Representative search/candidate/group/reconciliation tracing is reported in the next section, with unobserved stages retained explicitly. The initial HMM search is on in both HMM methods; this observation test is not an initial-HMM-off causal comparison. All 80 scores, 240 strata and 15 comparisons, full-precision stratum table, result interpretation.

Retained Fragment Pipeline Observations

The table counts the fixed representatives (HS 12, PHY 18, OF 11, OF checkpoint 12) in each arm, including errors, recoveries and retained TP controls. These are not rates, prevalence estimates or new accuracy endpoints. All 53 cases and empty bins remain available; no preferred stage explanation was used in selection.

Method	Recorded Decision	Baseline Representatives	Fragment Representatives
HS	group retention	9	6
HS	group separation	3	6
PHY	candidate separation	1	4
PHY	event rule retention	6	7
PHY	observed duplication	6	4

Method	Recorded Decision	Baseline Representatives	Fragment Representatives
	exclusion		
PHY	unambiguous bypass retention	5	3
OF	mcl group separation	0	2
OF	native pair retention	8	7
OF	within mcl native pair exclusion	3	2
OF checkpoint	mcl group separation	3	3
OF checkpoint	native pair retention	9	9

Three illustrative records show distinct observable boundaries, not causal or biological validation. HS Case0000 (unflagged FN) lacks direct significant hits in either direction, yet its graph is connected and its groups are separate. PHY Case0026 (two truncated endpoints, FN) has both directed significant hits and connected graph support, but its observed ancestor is called duplication and excludes the native pair. OF Case0030 (unflagged FN) is co-clustered at MCL but absent from native pairs; a matching event adapter is unavailable, so its tree-level cause is unknown.

Search-hit absence does not distinguish prefilter rejection from scoring or significance thresholds. Graph/group support is not orthology. A recorded duplication establishes the pipeline’s decision, not correctness of an inferred tree or the underlying evolutionary history.

Selection and empty bins, all retained observations, stage table, independent readback, actual execution and preserved refusal.

Gene-Tree And Event-History Controls Localize Errors

Across 10,125 fixed candidates, 1,377 received eligible gene-tree controls; 8,748 bypasses remained unchanged. Among eligible controls, 300 differed in unrooted topology and 30 additional candidates only in root. Generating-root mean F1 changed by +0.136 to +0.734 percentage points across the seven conditions. These descriptive effects are not significance tests or evidence of independent improvement. In divergence and divergence/turnover, respectively 12,736 of 12,766 and 13,072 of 13,104 residual generating-root false negatives (99.765% and 99.756%) already cross candidate boundaries. Changing gene trees alone cannot recover them. All-condition oracle results.

Of those cross-candidate losses, 11,442 and 11,554 (89.840% and 88.387%) lie in different saved graph components. The remaining 1,294 and 1,518 are connected but separated by candidates, including 117 and 128 direct graph edges. Almost all component-separated losses lack significant initial hits in either orientation. This identifies retained stage boundaries, not a single cause: missing hits cannot isolate the prefilter, and paths may cross other families. Complete upstream trace.

The complete residual cohort contains 20 candidates: eight eligible controls and twelve bypasses. All 800 cross-species pair rows reproduce 380 TP, 292 TN, 62 FN and 66 FP. All 62 false negatives receive high-confidence speciation calls before unsupported satellite sources are detached; their root groups agree, but the post-constraint filter removes them. Of the false positives, 45 are bypass calls at true duplication

ancestors and 21 receive speciation calls without retained species overlap. All 66 have overlap evidence in their full parent generating family that is absent from their retained candidate. Thus a single retained copy per species is not sufficient to establish orthology. These selected-cohort counts do not replace whole-panel accuracy. Event-history residual results.

The trace supports candidate-retention and membership-filter explanations for these fixed simulation cases. It does not establish that disabling constraints, merging candidates indiscriminately, or changing a search threshold improves independent real-data inference. All negative and neutral findings, bypasses and upstream losses remain. QfO/OrthoBench retain their primary role; these privileged-information simulations are mechanistic support, not substitutes for curated-reference validation or unfinished biological error strata.

Local Parameter Sensitivity Did Not Establish Improvement

The complete seven-arm panel includes separately admitted private-runtime high-CPM recovery. Its SwissTrees F1 was 83.253016% versus 83.351322% for the frozen control, a difference of -0.098306 percentage points with adjusted interval [-1.050447, 0.747525]. All 18 adjusted intervals include zero; no statistically supported parameter improvement is established. Identical observed normalization-arm statistics do not imply identical whole-proteome predictions or equivalence on unseen families. No defaults were changed. The all-arm figure retains every contrast and the full adjustment denominator.

Historical SIGSEGV, allocator and admission failures remain failed. Recovery followed a validated content-equivalent private control and has separate inference, conversion and score evidence; it neither repairs those earlier attempts nor proves their cause or memory safety. This panel is not an OrthoFinder superiority test or a controlled runtime comparison.

Transfer And Biological Recovery Reveal Trade-Offs

Frozen YGOB group-recovery F1 was 92.233654% for phylogenetic OrthoHMM and 92.318524% for full OrthoFinder. Their difference was -0.084870 points, with adjusted interval [-0.622528, 0.445222]. OrthoHMM had higher precision and lower recall. The evaluation projects predictions onto the reference universe and assumes exchangeable pillars; it does not establish unrestricted generalization or resolved pairwise orthology. YGOB results.

A descriptive partition by the frozen overlap screen retained all original false-positive allocations. In screen-negative pillars, phylogenetic OrthoHMM versus full OrthoFinder had F1 of 57.93% versus 45.02%, precision of 55.92% versus 30.74%, and recall of 60.10% versus 84.03%. In screen-positive pillars, their F1 values were 93.37% versus 95.01%. The negative stratum contained 2,893 singleton pillars out of 3,298, compared with 2,017 out of 6,952 in the positive stratum. This composition difference and the screen’s inability to exclude remote homology prevent independent-family or causal interpretations. The all-method stratum figure is descriptive, with no new uncertainty estimate or method tuning.

The prespecified whole-genome-duplicate application used experimental evidence from Kuzmin and colleagues (Kuzmin et al. 2020). All 240 experimental pairs were retained, with 239 input-eligible and 231 shared-reference-pillar pairs. On this development-exposed application, phylogenetic OrthoHMM separated 238 of 239 input-eligible pairs, versus 58 for high sensitivity and 236 for full OrthoFinder. Among the 231 reference-eligible pairs, however, only 193 OrthoHMM separations retained at least one non-S. cerevisiae reference homolog with each anchor, versus 56 for high sensitivity and 227 for full OrthoFinder. Mean per-pair homolog coverage was 82.338%, 99.149% and 98.413%, respectively.

Coverage counts reference homologs in the union of the anchor groups and can be high even when the anchors are merged; it is not orthology recall. The complete five-method comparison also retains SonicParanoid and the diagnostic OrthoFinder MCL checkpoint.

Phylogenetic OrthoHMM minus full OrthoFinder had a supported-separation difference of -14.719 percentage points, with adjusted interval [-21.645, -8.225], and a coverage difference of -16.075 points [-19.755, -12.496]. These exploratory percentile intervals use 20,000 paired pillar resamples and a 12-endpoint adjustment. Each eligible pair occupies a distinct pillar; these conditional estimates do not establish independent generalization. Supported separation does not establish cross-species copy-specific orthology. In the six prespecified case traces, five focal homologs left their anchor groups during root-lineage reconstruction, before satellite constraints. They remained in other output groups; no example was replaced to improve the result.

A subsequent fixed-tree diagnostic evaluated all four existing root-duplication rules on the same seven candidate families and six examples, with its protocol frozen before alternative outcomes were examined. The supported-children and confidence rules reproduced the species-overlap baseline partitions exactly. The mapped-event rule reduced homolog coverage in four of five reference-eligible examples, without improving supported separation. None recovered the five focal homologs into either anchor group. Native pair predictions and confidence annotations remained unchanged. This post hoc intervention rejects these three alternatives as a repair on the fixed inputs, not as methods in general. It neither establishes topology error nor identifies ancestral-copy truth, and no defaults were changed.

Shared-Host Resource Panel

The snapshot retains 27 of 27 reviewed attempts, 25 with measured native resources and 24 eligible observations. 6 method/size cells have three eligible repeats.

These are shared-host matched-resource observations with 32 native CPUs and a 128-GiB limit per run, not isolated comparative timing. Contention distortion is unknown and potentially method dependent. No background overhead is subtracted.

Tables report median [minimum, maximum] only for cells with three eligible repeats. An unavailable summary is not zero or a failed native inference. Ranges are not confidence intervals. Eligibility counts retain measured exclusions and pre-native aborts rather than selecting the fastest attempts.

Native command wall (seconds)

Method	Proteomes	Eligible/planned	Median [minimum, maximum]
OrthoHMM high sensitivity	4	2/3	Unavailable
OrthoHMM high sensitivity	8	3/3	1238.9352 [1221.4444, 1303.6330]
OrthoHMM high sensitivity	12	3/3	2720.0774 [2621.4976, 3027.0605]
OrthoHMM inferred phylogeny	4	2/3	Unavailable

Method	Proteomes	Eligible/planned	Median [minimum, maximum]
OrthoHMM inferred phylogeny	8	3/3	1999.5508 [1888.8847, 2086.9473]
OrthoHMM inferred phylogeny	12	3/3	4392.8716 [4042.4262, 4809.2924]
OrthoFinder 3.1.5 full	4	3/3	441.2890 [420.7400, 474.2999]
OrthoFinder 3.1.5 full	8	3/3	1139.9398 [1120.2548, 1179.9703]
OrthoFinder 3.1.5 full	12	2/3	Unavailable

Task-subtree CPU bracket (CPU-seconds)

Method	Proteomes	Eligible/planned	Median [minimum, maximum]
OrthoHMM high sensitivity	4	2/3	Unavailable
OrthoHMM high sensitivity	8	3/3	34395.7481 [33878.9946, 36610.5696]
OrthoHMM high sensitivity	12	3/3	78335.8306 [75126.8588, 86415.3192]
OrthoHMM inferred phylogeny	4	2/3	Unavailable
OrthoHMM inferred phylogeny	8	3/3	56008.3471 [52911.7819, 58240.7234]
OrthoHMM inferred phylogeny	12	3/3	125264.7375 [116000.9964, 136293.3322]
OrthoFinder 3.1.5 full	4	3/3	4602.7585 [4455.3737, 5008.8970]
OrthoFinder 3.1.5 full	8	3/3	17734.4088 [17232.2843, 18787.3118]
OrthoFinder 3.1.5 full	12	2/3	Unavailable

Native-step lifetime peak (GiB)

Method	Proteomes	Eligible/planned	Median [minimum, maximum]
OrthoHMM high sensitivity	4	2/3	Unavailable

Method	Proteomes	Eligible/planned	Median [minimum, maximum]
OrthoHMM high sensitivity	8	3/3	7.724 [7.715, 7.932]
OrthoHMM high sensitivity	12	3/3	10.555 [10.533, 10.730]
OrthoHMM inferred phylogeny	4	2/3	Unavailable
OrthoHMM inferred phylogeny	8	3/3	7.681 [7.677, 7.702]
OrthoHMM inferred phylogeny	12	3/3	10.511 [10.503, 10.603]
OrthoFinder 3.1.5 full	4	3/3	6.775 [6.769, 6.928]
OrthoFinder 3.1.5 full	8	3/3	10.226 [10.215, 10.233]
OrthoFinder 3.1.5 full	12	2/3	Unavailable

Excluded attempt indices: 0, 17, 20. Pre-native abort indices: 17, 20; these have no native resource endpoints, not zero measurements. Unreviewed indices: none.

Across measured attempts, maximum observed foreign CPU demand ranges from 42.1424 to 72.8708 core equivalents. These are process-interval observations, not reservations or estimates of causal slowdown.

CPU includes the native task-subtree wrapper bracket. Peak memory includes the native-step launcher and is not pure algorithm RSS. Preparation, conversion and scoring are outside the native command timer. This single nested taxon series co-varies proteome count and taxon composition, not taxon-invariant scaling. Native output validation does not provide new prediction-accuracy evidence. Historical DGX and earlier shared-host timings are not pooled.

The first high-sensitivity attempt completed native inference but failed the original process-monitoring cadence criterion. It remains a measured exclusion; the prospectively tested scheduling repair does not convert it to a pass. Two further attempts aborted before native inference: an OrthoFinder/twelve-proteome repeat encountered a preparation hash race, and a phylogenetic OrthoHMM/four-proteome repeat exceeded the environmental-response deadline. Both retain null native endpoints and separately reviewed infrastructure repairs. None of these failures was retried to fill a cell or obtain faster timing. All 27 planned identities are terminal-reviewed, including exclusions; this is not 27 eligible measurements. The three incomplete cells have two eligible repeats each, not completed three-repeat summaries.

Machine-readable final table, all attempts, complete-cell summaries, final resource figure, final terminal review, generated section and provenance.

Retained Development Exposure And Candidate Reproducibility

The family-evidence inventory contains 134 OrthoBench score blocks across 84 files and 77 SwissTrees blocks across 16 files: 7,770 and 1,386 family-block associations respectively. All 88 canonical families have explicit score evidence. Each of the 35 originally validation-labelled RefOGs appears in 18 declared validation blocks; each original development RefOG appears in 28 development blocks, and all 70 appear in 21 all-partition blocks. The historical 35/35 split remains recorded chronology, not untouched publication validation. Distinct count vectors number 84 and 36; neither different nor equal vectors establish independence or duplicate execution. The inventory also retains 34 subset-ordinal candidate-diagnostic containers and two synthetic scored-label containers without misassigning biological identities. Verified inventory.

Each original/native candidate trace contains 8,440 accepted merges. The three native repeats share 8,428, 8,439 and 8,426 semantic records with the original. Common cluster indices and hit counts agree; the maximum common support difference is 2.1316282072803006e-14. Every common anchor with changed selected source memberships reaches the four-attachment cap in both traces. In the frozen 19-gene fixture, baseline, order-reversed and one-unit-perturbed cases leave satellite IDs 8, 0 and 7 unattached, respectively: one unattached satellite and eight accepted merges in every case. These IDs are not numbers of missing satellites. This demonstrates that capped selection and tied/near-tied evidence can change assignments without changing merge counts. It does not establish historical relabeling, score-bit provenance, gene-tree correctness or an accuracy advantage. Complete diagnostic.

Historical Costs Remain Scope-Specific

A retained native-cost linkage checks six completed twelve-proteome Threadripper repeats against the original OrthoBench inputs and prescribed settings. All three high-sensitivity final partitions equal the original `p1_c0_r0` partition. The first phylogenetic final partition matches the original `p1_c1_r1`; the other two have changed groups containing 149 and 172 genes. All three candidate partitions differ, involving 70, 160 and 253 genes in affected groups. No changed group intersects a reference family. Affected-group gene counts are not counts of genes moved, and structural agreement is not new accuracy scoring or equivalence of all intermediate artifacts. Native wall medians are 45.335 and 73.215 minutes; these reuse the shared-host panel rather than provide new measurements or costs of the original cached executions. The patched native runtime differs from that deployment. Native cost/partition linkage.

The selected corrected QfO high-sensitivity and phylogenetic rows bind to `p1_c0_r0` group-derived pairs and `p1_c1_r1` native pairs respectively. Their retained resource associations describe one shared cached-replay worker (2,987.81 seconds), two preparation arms (3.035759820 and 77.641008615 seconds) and native reconciliation (6,822.365357 seconds). The repeated replay value is one observation, not two repeats; process RSS, sampled tree RSS and lifetime cgroup peak retain different meanings. Neither stage sums nor these native configuration observations fill the sixteen unavailable original factorial full-pipeline costs. The selected rows' complete cached-execution costs remain unknown. Exact QfO stage chains, factorial stage costs.

The additive all-tool metadata register retains 24 method/dataset rows and all 72 scientific score positions, with explicit conversion semantics and historical resource supplements. Selected QfO stage associations are added without replacing prior intervals; table entries are not independent timed experiments. Input consumption, transitive runtime attestation and missing whole-pipeline costs remain separately identified rather than inferred. All-tool metadata integration.

Discussion And Limitations

The supported contribution is an HMM-centered alternative with measurable component effects and explicit accuracy trade-offs. Initial HMM search has bounded support in matched-recall simulations; additional profile refinement has not demonstrated a general benefit. Neither simulation evidence nor an OrthoBench point advantage establishes superiority over full OrthoFinder.

The approximate significance formula is not universally calibrated under the tested synthetic conditions. The null experiment does not measure real-data orthology false-positive rates or causally explain any benchmark difference; independent calibration would be needed before claiming statistical significance for arbitrary compositions, search spaces, profiles or band settings.

The new error-stratum analyses describe development-exposed families, not independent confirmation. Fixed-model branch-path distances are an explicit estimated sequence-divergence descriptor, not calibrated biological time, known ancestral history, verified model adequacy or a causal explanation. Paralogy, sampling, composition, domains/length and gaps/alignment error may affect them. Independent arithmetic does not eliminate shared parsing, alignment/model assumptions or heuristic topology uncertainty. Length and composition remain proxies rather than literal fragment truth; Pfam summaries are not complete validated architectures, and informative reference-node duplication fractions are not complete ancestral histories. None of these results promotes a new default or closes the remaining descriptor and generalization gaps.

Original TreeFam-A family mappings and complete source trees remain unavailable. Public QfO container searches now cover recognized benchmark build contexts for all 40 listed tag names at retained digests, without finding original NHX/mapping filename candidates. Other layers and archive-embedded or differently named contents remain outside that search; no family labels are inferred from its negative result. Public archive recovery yielded historical Selectome subtrees, but all are restricted to Euteleostomi. A taxonomic coverage audit found that 55,933 of 79,320 retained reference relations (70.52%) have at least one endpoint outside that clade. Those relations cannot be reconstructed from these subtrees under a species-consistent mapping; the other 23,387 relations are only potentially in scope, not demonstrated reconstructions. This is a coverage exclusion, not an accuracy effect or a basis for family-level intervals. TreeFam is retained as descriptive supporting evidence, without inventing family labels or TreeFam-specific significance/generalization claims. The recorded search scope is finite, not proof of global unavailability; further public searches require a concrete new lead. Neither recovery of the originals nor TreeFam family-level intervals is a completion prerequisite under the approved reporting scope.

VGNC dependence and rare-error diagnostics do not justify the failed candidate confidence-interval procedure. The conventional native VGNC and GO/EC half-widths shown below remain available, with their actual formulas and denominators, but do not establish dependence-aware paired differences. FAS biological/generalization and secondary-mean uncertainty are unavailable. These limitations remain visible and nonblocking: no new intervals are invented, no endpoint is changed, and inferential claims are restricted to analyses with supported uncertainty. The six-metric mean is descriptive, not an official overall accuracy measure or a basis for choosing a winner. The corrected FAS sample audit reproduces all eight means, but scored fractions range from 0.0067% to 58.46% of reported eligible pairs and every sample reuses proteins across pairs. Native pair-level standard errors do not resolve this comparison uncertainty. The requested-score audit found 1-49 missing new scores per method except OrthoMCL, which lacked 1,252 of 9,000. Assuming omitted scores lie in [0,1], its intended-sample mean is bounded by 0.724872-0.736975; this is not a confidence interval. A native-complexity audit found no above-limit protein in saved new-score pairs, but found them in precomputed pairs. This agrees with the demonstrated cutoff mechanism without identifying historical omitted pairs or establishing comparison bias. The completed eligible-population audit matches all eight methods' native logged counts, reusing six fixed recounts and freshly scanning two. Pairs absent from the precomputed lookup constitute 5.13% of satellite_v2's eligible set, 16.86% of full OrthoFinder's and 82.05% of its sequence-only checkpoint's. Assuming their scores lie in [0,1] gives conservative full-mean bounds of 0.733966-0.785282 for satellite_v2 and 0.605946-0.774511 for full OrthoFinder. These overlapping bounds neither establish a full-eligible-set advantage nor replace native sample scores or confidence intervals. Some lookup-missing pairs were newly scored in native samples; those values are not used to tighten these bounds. An eight-method stratum-weight decomposition separates count rounding from omission-induced changes in the saved mixture. Holding the observed stratum means fixed, native-minus-reweighted-saved-strata differences are +0.001318947 FAS units for OrthoMCL and +0.000030716 for phylogenetic OrthoHMM. These are numerical diagnostics, not corrected benchmark scores or estimates of population bias. Omissions can also change the stratum means; the unknown omitted pair identities, unseeded selection

and dependence remain unresolved. No FAS confidence interval or new method ranking follows. The conditional repeated-statistic analysis above adds a distinct sampling target for the retained eight-method FAS comparison. It does not identify full-eligible-population functional accuracy or new-family/clade uncertainty. Earlier population-bound and stratum-weight diagnostics alone do not supply confidence intervals; the conditional ranges additionally require the stated uniform/fixed-return design assumptions and validation. Historical source identity and biological/generalization gaps remain unresolved. The newer native factorial's GO/EC/FAS uncertainty is not supplied by this older snapshot.

The completed simulations and tree perturbations do not cover arbitrary evolutionary conditions, and novel-taxon YGOB testing retains homolog-family overlap with development data. Generating-tree, upstream and residual diagnostics localize errors in the retained development-exposed panel but do not establish empirical gene-tree accuracy, ancestral-copy root-HOG truth or independent generalization. The development inventory establishes retained evaluation exposure, not a complete causal tuning history, all historical experiments or family-disjoint validation. Its frozen local input set must be supplied to reproduce that expanded inventory. Capped tie-sensitive candidate selection is a demonstrated reproducibility limitation, but does not justify benchmark-chosen score rounding or attributing changes to competing workloads. A method change would require a new freeze and independent confirmation. Candidate retention can remove species-overlap evidence, and strict satellite detachment can remove otherwise correct pair calls. The demonstrated cases motivate testable hypotheses, not unvalidated new defaults.

Recovered OrthoMCL results retain sequence-specific BLAST failures. The separately recovered high-CPM result does not replace failed historical runs; no missing score was imputed and no failure was recategorized as a success. Shared-host runs and the historical DGX panel are descriptive resource records. The replacement matched-resource Threadripper panel has completed under the shared-host amendment, not as an isolation-controlled experiment. Competing workloads may affect tools differently, and identical resource limits do not identify isolated speed or causal speedups. The final figure retains excluded measurements and missing eligible repeats; it cannot support an isolated efficiency ranking. Six cells have three eligible repeats, while three cells retain unavailable summaries. This single nested taxon series does not isolate taxon-count scaling from proteome composition. The versioned local rc6 archive supports its documented selected-byte/reporting restoration scope; it is not a public deposition or proof of whole-study native restoration. Redistribution limitations and journal-specific formatting remain explicit. Completed resource reporting does not remove scientific evidence limitations, and this revision's descriptive/conditional uncertainty scope does not require resolving those gaps to report bounded publication claims.

Conditional Count Model Does Not Establish Native Uncertainty

A restricted mathematical experiment examines a fixed perfect-recall truth mass and Poisson shared/method-only whole-panel false-positive counts. Its target is the difference of F1 ratios at expected count vectors, not expected sample F1 or generalization from native families. Simultaneous component Poisson mean limits are sharply projected through that ratio difference. Zero observed errors retain a positive rate bound, rather than a zero-width plug-in interval. The model is not fitted to native errors.

All 12 prespecified independent-Poisson enumeration cells passed; minimum covered probability mass was 0.999559608605, with maximum omitted tail 5.999645225074e-13. This finite panel is conservative, not proof of a native law or floating-point certification. Both non-Poisson common-shock controls had zero coverage despite having the same mean-count target form. Imperfect recall, unequal family sizes and shared-clade dependence remain outside the construction. Earlier failed dyadic screens are not replaced.

No native VGNC, TreeFam, GO/EC/FAS or secondary-mean confidence interval is admitted, and no benchmark statistic, resampling unit or default is changed. Frozen model and derivation, all results and failure controls, execution and independent tail/corner readback.

Controlled Fragment Scope And Limitations

The controlled observation result supports one bounded synthetic-fragment diagnostic, not a claim of better natural-fragment recovery or broad superiority to full OrthoFinder. It preserves the negative comparisons. Development exposure, artificial centered truncation, absence of validated domain/indel truth, ten-seed approximate interval tails and the changed full OrthoHMM runtime inventory limit interpretation. Counts that change in the unflagged stratum do not identify which search, grouping or reconciliation step caused the error. No initial-HMM-off control is supplied by this extension.

The fragment job is not a replacement scaling panel. Cached baseline costs are not paired timings, and the checkpoint has no independent runtime. Timing measurements were collected on a shared Threadripper while other analyses were running. Competition for CPU, memory bandwidth and I/O may have affected elapsed times, with an unknown and potentially tool-dependent impact. These are observed shared-host timings, not estimates of isolated performance. This extension does not resolve unavailable original TreeFam resources, failed native factorial cells or other unmet publication requirements.

Representative Trace Limitations

Minimum-hash retrospective examples are not a random family sample or new independent generalization test. Category overlap is allowed; realized unique cases do not remove dependence. Selected counts cannot quantify the prevalence or causal contribution of search, clustering or reconciliation errors. Native node calls do not validate inferred trees against biology. Unavailable OF search/event evidence remains null, not a negative result. This diagnostic does not establish natural-fragment recovery, initial-HMM benefit, a default parameter improvement, isolated timing performance or publication readiness. The full benchmark uncertainty, failed-cell and shared-host limitations remain.

Reproducibility And Availability

The extended manuscript links frozen protocols, source revisions, seeds, input/output manifests, conversion audits, executable workflows and generated figures. Installed inference and independent readback have been validated on full OrthoBench, but that is not proof of all-method cross-host portability. Historical locks are retained as provenance; patched replacement environments must be distinguished from the binaries used for original scores. The QfO parameter numerical component passed fresh local archive extraction, copied byte verification and reproduction of all 18 endpoints within 1e-12 using unchanged arithmetic. A Python-event guard rejected its original-checkout canary and observed no later original-path events or project-module imports. This is not OS containment, a new independent statistical implementation, raw-reference recount, native inference restoration, cross-host validation or redistribution clearance.

A separate descriptive-table component restored under standard-library Python 3.12 and reproduced all 208 rows and 984 score/difference cells across four SwissTrees tables and eight methods. This verifies arithmetic from retained sufficient statistics and annotations, not raw-source admission, new benchmark scoring or bootstrap uncertainty. An input-only private archive workflow then restored the complete duplication and fragment dependency panels, retaining all 11 and 1,774 original record occurrences

respectively. Independently pinned archive and binding digests preserve source identities at new locations. All 27 affected regression/options cases passed using the restored inputs, including four raw-export regressions with unchanged assertions. The fresh children blocked original-checkout reads and subprocesses; these Python-event guards are not OS containment. Native annotation extraction/admission was not rerun, and private archives remain unuploaded with redistribution uncleared. Neither component establishes complete executable study restoration or all-method cross-host portability.

The null-score observations retain all 180,000 synthetic score evaluations and their source/runtime pins. Its public numerical readback checks all 90 tail endpoints without loading the native kernel; this is not a complete inference runtime archive.

The standalone YGOB arithmetic replay uses identifier-free retained counts to reproduce all 12 point metrics and six nominal/adjusted interval endpoints after local archive restoration. This is not native re-admission, proof of pillar exchangeability or independent family validation; original transfer scores and overlap limitations remain.

The standalone simulation replay reproduces retained-count arithmetic for the two length panels without historical file reads. It preserves fixed-length unavailable comparisons and variable-length conditional effects; native admission and tree-control inference are not rerun.

The final resource-reporting archive restores outside the repository and reproduces all three table files, PNG pixels and resource-section text exactly from its copied isolated reader. It includes all attempts and failures without opening original evidence paths. This is reporting portability, not raw-accounting or native reproduction, isolated timing, complete study restoration or redistribution clearance. The source/review export integration adds explicit result-helper and final-manuscript selection; fixture tests do not establish an actual full-study archive.

The progress ledger records completed work and unmet requirements. No submission-ready release or archival DOI is claimed. Dated main-text HTML/PDF and review archives preserve the exact inputs of their own snapshots; older versions are not rendered copies of revised Markdown. A refreshed review is not a complete executable study release. This revision incorporates the completed shared-host resource panel while also integrating the later simulation mechanism diagnostics and preserving scientific limitations. The three hash-pinned checked-summary reports support a standalone reporting replay using standard-library Python; it recomputes the finite-panel means, stage-count identities and residual classifications, not native search/inference or original XML admission. Its separate execution and package receipts govern what has actually been restored and reproduced. The second revision’s 34-page review/rc3 archive, third revision’s 35-page review/rc4 archive and first 6 October direct-review component remain historical. They do not contain this later four-cell native QfO source snapshot. Fresh native tables, bounded interval bindings, existing figures and pair diagnostics retain their source identities and failures. This source is mechanically extended from already-reviewed summaries, without new scoring, bootstrap draws, admission, inference, default selection or independent-validation claims. Any new local render/review or direct-link package includes only the content and checks in its own dated receipt. Such reporting portability does not supply transitive raw/dependency closure, a hermetic full-study reproduction, redistribution clearance, public deposition or an archival DOI. Existing replay receipts remain valid only within their recorded scopes.

The new three-cell fixed-bin figure and model-distance supplements retain all positive, negative, neutral and empty outcomes. A small model-distance evidence archive contains all 18 selected alignments and the complete new tree/log/pair/receipt payloads; execution receipt and independent readers state the exact scope. Auxiliary inference Slurm TotalCPU and MaxRSS were unavailable, not measured zeros or native tool timing. The IQ-TREE 3 citation uses retained publisher-deposited Crossref metadata for the current

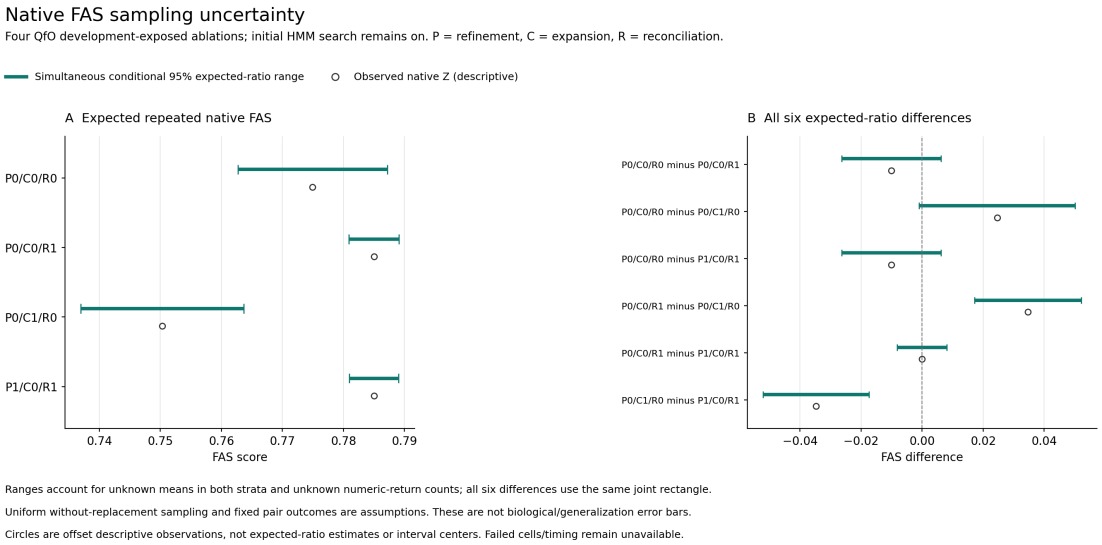
Wong et al. 2026 paper, rather than the installed binary's old submitted-paper notice. The executed binary remains 3.0.1; no software update or rerun follows. The new bibliography preserves every prior entry and adds this reference only. Earlier manuscript reviews and rc5 remain unchanged; their dated receipts do not establish review or package inclusion of this later source revision. This is reporting integration, not transitive raw/dependency closure, a hermetic whole-study release, redistribution clearance, independent confirmation or DOI deposition. Future render/package receipts must cover their actual new inputs.

Relocated Fragment Raw Stage Replay

The selected fragment-stage postprocessing was executed from a separately located component containing 1,783 pinned artifact files (39,428,185 payload bytes) and unchanged native-reader kernels. Its complete result matches the retained native readback: 106 observations, 53 selected cases, 44 contexts and 415 original input references. The first ownership-map failure is preserved; the corrected map includes all configured ownership FASTAs. Python-open auditing observed no original-artifact fallback, not operating-system containment. This is raw-stage postprocessing portability, not a repeat of historical inference, new biological evidence or whole-study restoration.

The component remains local. The Git repository contains its map, result and reproduction instructions, not the raw payload; no public deposition or redistribution clearance is inferred. The existing 30-page review PDF remains the rendered copy of the prior manuscript, not this extended text. Replay result and preserved failure, complete copied readback, reproduction workflow.

Conditional Native FAS Sampling Figure



Conditional native FAS ranges and all six differences

Figure. Native FAS sampling uncertainty in four admitted QfO ablations. Panel A shows simultaneous conditional 95% ranges for expected repeated native post-attrition FAS; offset open circles show the actual descriptive native Z. Panel B shows all six expected-ratio difference ranges; its offset circles show observed differences of Z, not expected-difference point estimates. Bars have no fitted central estimate. The dashed line marks zero. Four difference ranges include zero, including the reconciliation and additional profile-refinement contrasts; inclusion is not equivalence. Two ranges exclude zero for R-on versus candidate-expanded R-off, without isolating a single-component causal effect. Labels preserve original contrast orientation.

The same joint rectangle covers four cells and six contrasts, using three components per cell with error 0.05/12 each. Both finite-population stratum means and the number of numeric-return missing pairs are unknown. The construction assumes uniform without-replacement within-cell sampling and fixed pair outcomes, not biological pair/family IID, cross-cell independence or missing-at-random outcomes. This is conditional sampling uncertainty in development-exposed QfO data, not an interval centered on observed Z, biological generalization or overall accuracy superiority. Initial HMM search stays on; P denotes profile refinement, C candidate expansion, R reconciliation. Three failed native cells and the failed P0/C0/R1 timing remain unavailable. No inference/scoring/sample was rerun.

The figure source was committed/pushed before production rendering. All 19 focused tests passed; actual PNG, SVG and PDF rendering completed once. Scoped readback verifies all four original rows and six contrasts, four zero-overlap ranges, output hashes, vector labels and 2900x1360 raster dimensions with 147168 nonwhite pixels. Actual visual inspection found no overlapping or cropped labels/captions. This does not render or certify the complete manuscript.

Editable SVG, vector PDF, plotted values and provenance, actual render/readback.

Conventional Native VGNC And GO/EC Bars

These tables use the corrected September eight-method snapshot, not the newer four-cell ablation. HS means high sensitivity; PHY means satellite_v2. The OrthoFinder checkpoint is a diagnostic pre-phylogenetic output, not a completed alternative phylogenetic pipeline. FastOMA used a supplied OrthoFinder species tree; OrthoMCL retains recovered BLAST failures. All values are unchanged native aggregates, rounded here to six decimals. A displayed +/- quantity is the native approximate 95% half-width, not one SEM, a paired difference interval or an F1 interval. Bars do not correct shared-protein/family dependence. Overlap/non-overlap is not used as a paired significance test.

Table. VGNC recall and precision with conventional binomial half-widths. All methods use 23,934 asserted reference pairs for the recall denominator; TP+FP is the method-specific precision denominator. It excludes unscored relations under the conservative native negative-reference rule.

Method	Recall +/- Native Half-Width	Precision +/- Native Half-Width	TP + FP
OrthoHMM HS	0.834503 +/- 0.004708	0.554021 +/- 0.005131	36051
OrthoHMM PHY	0.821426 +/- 0.004852	0.999339 +/- 0.000359	19673
OrthoFinder full	0.982661 +/- 0.001654	0.994503 +/- 0.000942	23649
OrthoFinder checkpoint	0.995362 +/- 0.000861	0.076891 +/- 0.000938	309827
SonicParanoid	0.971338 +/- 0.002114	0.994524 +/- 0.000946	23376
ProteinOrtho	0.930559 +/- 0.003221	0.980541 +/- 0.001796	22714
FastOMA	0.908415 +/- 0.003654	0.995786 +/- 0.000859	21834

Method	Recall +/- Native Half-Width	Precision +/- Native Half-Width	TP + FP
OrthoMCL	0.982702 +/- 0.001652	0.476750 +/- 0.004407	49334

Table. GO/EC mean functional similarity with native Darwin Student-t half-widths. Counts are eligible scored pairs, not all predictions or unique proteins. Different methods evaluate different pair sets; higher means do not alone establish better ortholog recovery. The native half-width is InverseStudent_t(0.95,n-1)*sqrt(sample_variance/n), computed before raw serialization. These quantities are not independent biological intervals.

Method	GO Mean +/- Native Half-Width	GO Pairs	EC Mean +/- Native Half-Width	EC Pairs
OrthoHMM HS	0.472271 +/- 0.001198	145619	0.932038 +/- 0.000832	185664
OrthoHMM PHY	0.490349 +/- 0.001600	84211	0.965650 +/- 0.000722	117460
OrthoFinder full	0.469548 +/- 0.001128	163557	0.936130 +/- 0.000846	175361
OrthoFinder checkpoint	0.407043 +/- 0.000289	1979084	0.753642 +/- 0.000463	1660940
SonicParanoid	0.454385 +/- 0.001017	197930	0.872761 +/- 0.000964	244356
ProteinOrtho	0.486336 +/- 0.001524	89609	0.963168 +/- 0.000661	143135
FastOMA	0.436894 +/- 0.001093	171132	0.897168 +/- 0.001104	178598
OrthoMCL	0.463755 +/- 0.001083	175181	0.921326 +/- 0.000829	220622

The 24 source aggregate hashes were checked against their retained admissions; axes, counts and values match the existing corrected comparison. Native participant IDs agree within each method's three admission-bound aggregates. No normalization, identity replacement or score change was used. VGNC half-width formulas reproduce all eight retained pairs of bars with the native integer denominators. Existing GO/EC runtime/formula evidence is reused, not rerun or relabeled as a new statistical validation. Original corrected scores and semantics, retained admission register, GO/EC runtime and arithmetic evidence.

References

The bibliography attributes the methods and resources cited above. Full execution evidence and additional dependency references remain in the extended manuscript; this selected list is not complete software attribution.

Altenhoff, Adrian M, Javier Garrayo-Ventas, Salvatore Cosentino, David Emms, Natasha M Glover, Ana Hernández-Plaza, Yannis Nevers, et al. 2020. “The Quest for Orthologs Benchmark Service and Consensus Calls in 2020.” *Nucleic Acids Research* 48 (W1): W538–45. https://doi.org/10.1093/nar/gkaa308.

Buchfink, Benjamin, Klaus Reuter, and Hajk-Georg Drost. 2021. “Sensitive Protein Alignments at Tree-of-Life Scale Using DIAMOND.” *Nature Methods* 18 (4): 366–68. https://doi.org/10.1038/s41592-021-01101-x.

Byrne, Kevin P., and Kenneth H. Wolfe. 2005. “The Yeast Gene Order Browser: Combining Curated Homology and Syntenic Context Reveals Gene Fate in Polyploid Species.” *Genome Research* 15 (10): 1456–61. https://doi.org/10.1101/gr.3672305.

Cosentino, Salvatore, Sira Sriswasdi, and Wataru Iwasaki. 2024. “SonicParanoid2: Fast, Accurate, and Comprehensive Orthology Inference with Machine Learning and Language Models.” *Genome Biology* 25 (1). https://doi.org/10.1186/s13059-024-03298-4.

Davín, Adrián A, Théo Tricou, Eric Tannier, Damien M de Vienne, and Gergely J Szöllősi. 2019. “Zombi: A Phylogenetic Simulator of Trees, Genomes and Sequences That Accounts for Dead Linages.” *Bioinformatics* 36 (4): 1286–88. https://doi.org/10.1093/bioinformatics/btz710.

Emms, David M, and Steven Kelly. 2020. “Benchmarking Orthogroup Inference Accuracy: Revisiting Orthobench.” *Genome Biology and Evolution* 12 (12): 2258–66. https://doi.org/10.1093/gbe/evaa211.

Emms, David M., Yi Liu, Laurence Belcher, Jonathan Holmes, and Steven Kelly. 2026a. “OrthoFinder: Improved Phylogenetic Orthology Inference with Enhanced Accuracy and Scalability.” *Nature Methods* 23 (7): 1327–33. https://doi.org/10.1038/s41592-026-03126-6.

———. 2026b. “Author Correction: OrthoFinder: Improved Phylogenetic Orthology Inference with Enhanced Accuracy and Scalability.” *Nature Methods*, August. https://doi.org/10.1038/s41592-026-03238-z.

Klemm, Paul, Peter F. Stadler, and Marcus Lechner. 2023. “Proteinortho6: Pseudo-Reciprocal Best Alignment Heuristic for Graph-Based Detection of (Co-)orthologs.” *Frontiers in Bioinformatics* 3 (December). https://doi.org/10.3389/fbinf.2023.1322477.

Kuzmin, Elena, Benjamin VanderSluis, Alex N. Nguyen Ba, Wen Wang, Elizabeth N. Koch, Matej Usaj, Anton Khmelinskii, et al. 2020. “Exploring Whole-Genome Duplicate Gene Retention with Complex Genetic Interaction Analysis.” *Science* 368 (6498). https://doi.org/10.1126/science.aaz5667.

Li, Li, Christian J. Stoeckert Jr., and David S. Roos. 2003. “OrthoMCL: Identification of Ortholog Groups for Eukaryotic Genomes.” *Genome Research* 13 (9): 2178–89. https://doi.org/10.1101/gr.1224503.

Majidian, Sina, Yannis Nevers, Ali Yazdizadeh Kharrazi, Alex Warwick Vesztrocy, Stefano Pascarelli, David Moi, Natasha Glover, Adrian M. Altenhoff, and Christophe Dessimoz. 2025. “Orthology Inference at Scale with FastOMA.” *Nature Methods* 22 (2): 269–72. https://doi.org/10.1038/s41592-024-02552-8.

Manni, Mosè, Matthew R Berkeley, Mathieu Seppey, Felipe A Simão, and Evgeny M Zdobnov. 2021. “BUSCO Update: Novel and Streamlined Workflows Along with Broader and Deeper Phylogenetic Coverage for Scoring of Eukaryotic, Prokaryotic, and Viral Genomes.” *Molecular Biology and Evolution* 38 (10): 4647–54. https://doi.org/10.1093/molbev/msab199.

Quest for Orthologs consortium, Adrian M Altenhoff, Brigitte Boeckmann, Salvador Capella-Gutierrez, Daniel A Dalquen, Todd DeLuca, Kristoffer Forslund, et al. 2016. “Standardized Benchmarking in the Quest for Orthologs.” *Nature Methods* 13 (5): 425–30. https://doi.org/10.1038/nmeth.3830.

Spielman, Stephanie J., and Claus O. Wilke. 2015. “Pyvolve: A Flexible Python Module for Simulating Sequences Along Phylogenies.” *PLOS ONE* 10 (9): e0139047. https://doi.org/10.1371/journal.pone.0139047.

Steenwyk, Jacob L, Thomas J. Buida III, Antonis Rokas, and Nicole King. 2024. “OrthoHMM: Improved Inference of Ortholog Groups Using Hidden Markov Models.” Preprint. openRxiv. https://doi.org/10.1101/2024.12.07.627370.

Traag, V. A., P. Van Dooren, and Y. Nesterov. 2011. “Narrow Scope for Resolution-Limit-Free Community Detection.” *Physical Review E* 84 (1). https://doi.org/10.1103/PhysRevE.84.016114.

Traag, V. A., L. Waltman, and N. J. van Eck. 2019. “From Louvain to Leiden: Guaranteeing Well-Connected Communities.” *Scientific Reports* 9 (1). https://doi.org/10.1038/s41598-019-41695-z.

Wong, Thomas K F, Nhan Ly-Trong, Huaiyan Ren, Piyumal Demotte, Hector Baños, Andrew J Roger, Edward Susko, et al. 2026. “IQ-TREE 3: Phylogenomic Inference Software Using Complex Evolutionary Models.” *Molecular Biology and Evolution* 43 (5): msag117. https://doi.org/10.1093/molbev/msag117.